

RoySearch: Model Notes

Two-Ability Comparative Advantage in Segmented Search Markets

(Linear sectoral production, skill-copula correlation ρ_x , endogenous training and cross-market search)

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Scope. This document contains the full mathematical description of the *RoySearch* model: primitives, value functions, equilibrium conditions, solution algorithms, and transition dynamics. It is the companion reference for the paper and the codebase, and it supersedes the single-ability notes. The one structural change relative to the previous model is the worker type: a scalar innate ability x is replaced by a *pair* of abilities (a_U, a_S) — an aptitude for unskilled work and an aptitude for skilled work — drawn from a joint distribution with correlation ρ_x . Market separation is delivered by comparative advantage in the sense of Roy (1951), formalised through the copula/concordance apparatus of Gola (2021), rather than by curvature of the production technology. All search, bargaining, damage-shock, training, demographic, free-entry, and measurement machinery is retained. The previous one-dimensional model is the $\rho_x \rightarrow 1$ limit. A second, self-contained refinement of the skilled block separates the distribution from which new matches draw their quality (the *offer* distribution, on $[0, 1]$) from the distribution governing post-shock redraws (the *shock* distribution, compressed onto $[0, \delta]$); the gap between them, indexed by a single scale δ , is what activates the on-the-job-search ladder and endogenous skilled separation (Section 11). It nests the single-distribution skilled block as the $\delta \rightarrow 1$ case.

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1 Overview: what changes and why

The problem. In the previous model each worker carried a single innate type $x \in [0, 1]$, and both labour markets read that same index. Separation of the two markets — keeping both populated in equilibrium — had to be manufactured through the shape of the skilled production technology: the skilled loading was made *convex* (exponential, then x^2) while the unskilled loading was linear (x), so that the skilled/unskilled output ratio rose in x and generated single-crossing comparative advantage in one dimension. Estimation of that device was fragile: the productivity/gradient objects were pushed to corners or, in richer variants, into an economically indefensible region in which measured production fell in ability. A model in which a more able worker produces less cannot be defended, whatever objective value it attains.

The fix. Give each worker *two* abilities and let comparative advantage separate the markets. A worker who is good at unskilled work and bad at skilled work stays unskilled — not because unskilled work rewards low ability, but because that worker is genuinely better at it. Production is then linear and strictly increasing in *own* ability in both sectors; the burden of separation is carried by the joint distribution of (a_U, a_S) , summarised by their correlation ρ_x .

Why ρ_x separates the markets. This is the mechanism of Gola (2021), in which each sector uses a different coordinate of a worker’s skill vector and sorting is by comparative advantage. Gola’s central comparative static, proved via the concordance ordering of the skill copula (Scarsini, 1984), is that when the concordance (positive dependence) of skills is high a few workers are good at everything and the rest are good at nothing; when concordance is low (negative dependence) *most workers are good at exactly one job*, so each sector retains a large pool of workers suited to it. Negative correlation between a_U and a_S is therefore the force that keeps both markets populated at once — the property the single-ability model could not achieve. A minimal directed-search prototype confirms the mechanism survives search frictions and a participation margin: sweeping ρ_x at fixed productivity moves the unskilled market between a healthy share and near-collapse, with no wrong-signed production gradient anywhere.

Net parameter change. The skilled-loading curvature parameter is dropped in favour of a linear productivity in a_S ; a single skill-copula correlation ρ_x is added. In the symmetric-marginals baseline no new marginal parameters are introduced (both abilities share a_ℓ, b_ℓ). The change is thus a near-neutral swap: one shape device removed, one correlation added.

A second, self-contained fix: the skilled job ladder. Independently of the type-space change, the skilled block is refined so that new matches and post-shock redraws no longer read the same quality distribution. Offers are drawn on $[0, 1]$; shocks demote a match onto $[0, \delta]$, with $\delta \in (0, 1]$ the only new parameter. Having separate distributions for offers and shocks is Pissarides (2000, Ch. 4)’s device for placing new matches above the deteriorated ones, and it is what lets on-the-job search and endogenous separation switch on at all — both are identically zero when offer and shock coincide (older version of this model), because a shock is then no worse than a fresh hire. δ is the cheapest way to separate the distributions, since it allows me to separate the two distributions without having to introduce more parameters through an entirely new distribution. The refinement nests the old version ($\delta = 1$). The economics is developed in Section 11.

Separation in both markets: an exogenous hazard and an endogenous margin, separated by a second flow moment. Following the canonical Pissarides (2000) architecture, job destruction in each market has two components: an *exogenous* layoff hazard ξ_j (plant closures, firm-level demand shocks, reorganisations — events orthogonal to the idiosyncratic match quality), and an *endogenous* margin driven by the quality shock crossing the reservation, $\lambda_j G(p_j^*)$. Both are estimated inside the SMM loop. The realised separation hazard is their sum, $\xi_j + \lambda_j G(p_j^*)$, so the separation moment `sep_rate_j` alone does *not* separate them — both send the worker to unemployment, and because the quality shock is a memoryless redraw (Section 11), an exogenous and an endogenous separation are observationally identical *as separation events*: neither the pre-separation wage nor the tenure of the separator distinguishes them (the redraw carries no memory, so the separation hazard is flat in tenure and the pre-shock quality of a separator is a random draw from the surviving distribution under either channel). The split is nonetheless identified, because the redraw leaves a second, distinct footprint that ξ_j does not. **Every arrival of the quality shock produces one of two observable events: if the redraw falls below reservation ($p' < p_j^*$) the match is destroyed (an endogenous separation, rate $\lambda_j G(p_j^*)$); if it survives ($p' \geq p_j^*$) the worker keeps the job but the match quality — and hence the wage — jumps to the new draw (a within-job wage change, rate $\lambda_j [1 - G(p_j^*)]$).**

The exogenous hazard ξ_j generates only separations; it never churns a survivor’s wage. A *within-job wage-change hazard among job stayers* therefore loads on λ_j and is blind to ξ_j : it pins the shock arrival rate, and `sep_rate_j` then identifies ξ_j as the residual $\text{sep_rate}_j - \lambda_j G(p_j^*)$. The *endogenous separation share* $s_j^{\text{endo}} \equiv \lambda_j G(p_j^*) / (\xi_j + \lambda_j G(p_j^*)) = 1 - \xi_j / \text{sep_rate}_j$ is then a genuinely targeted object rather than a free split the optimiser can collapse to zero: the corner $p_j^* \rightarrow 0$ (a dead endogenous margin) forces the within-job wage-change hazard to its maximum, λ_j , so a finite measured `wchg_rate_j` below that ceiling is what disciplines the split away from the corner. The within-job wage-change moment is measured from the SIPP (data-and-moments companion); the unskilled market, which has no on-the-job search, gives the cleanest read (the survivor’s wage change is purely the redraw), while the skilled version additionally loads on the job ladder and is informative jointly with it.

One moment the ladder overshoots, and why it is structural rather than a tuning failure. The memoryless redraw governs the skilled job ladder, and it fits the flow of employer-to-employer moves — how *often* skilled workers switch jobs — but not their *size*. In the data a skilled worker who moves employer-to-employer gains a modest raise, about 11% in log wages (`ee_step_S` = 0.109, SIPP). The model produces a raise roughly ten times that. The reason is mechanical and unavoidable within the memoryless architecture: a mover draws his new match quality afresh from the *entire* offer distribution, uncorrelated with the job he is leaving, so the typical accepted offer sits far above his current match and the implied wage jump is enormous. This is not a corner the optimiser wandered into; it is what a memoryless redraw *is*. The natural first suspect, the bargaining share β_S , cannot repair it: β_S governs how the surplus of a given match is split, not how far apart successive matches lie, and sweeping it across its whole range leaves the step trapped near ten times the target at every value (Figure 2a). No amount of re-bargaining makes a random redraw a small step.

A different wage rule cannot fix it either: the Bertrand alternative. Before localising the redraw, it is worth asking whether a richer *wage* protocol repairs the step. Replace Nash splitting with Bertrand competition in the manner of the sequential-auction literature: when a poaching offer arrives, the incumbent and the entrant bid for the worker, and the winner pays up to the loser’s value, so the mover’s wage is bid up toward the full quality of the new match rather than settling on a share of its surplus. This is the strongest possible wage response to a move, and it pushes in the right direction — it cannot raise the step, only compress it. Yet it cannot compress it enough, and the reason is the same one that defeats β_S : the wage rule governs how the surplus of a poached match is *split*, while the overshoot is set by how far the new match *lies* from the old. Offers are still drawn globally, so a mover still lands far above his incumbent quality. Sweeping β_S across its range and computing the fully-bid-up Bertrand floor at each value — the smallest EE step this protocol could produce — the floor stays near 1.1 in log wages at the pinned $\beta_S = 0.272$ and falls only to about 0.98 at $\beta_S = 0.38$, never coming within an order of magnitude of the 0.109 target (Figure 1); the mean accepted draw behind it is $R \approx 11.7$, far above the incumbent match. Bertrand competition is a wage mechanism, and the overshoot is a mobility one: no rule for dividing a match’s surplus can make a globally-drawn offer a small step.

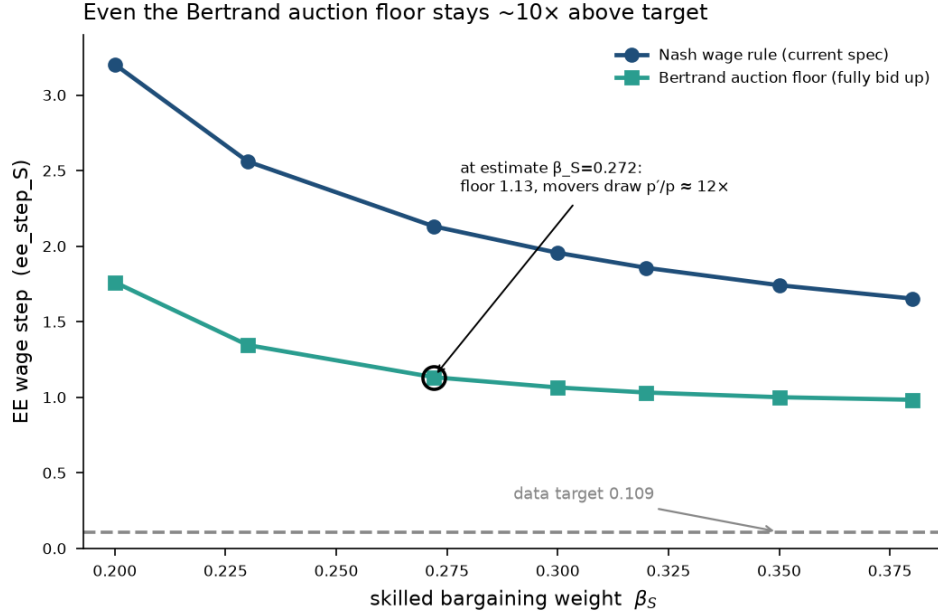


Figure 1: The skilled EE wage step under two wage protocols, swept over the bargaining share β_S : the Nash-bargained step (as shipped) and the fully-bid-up Bertrand floor (the smallest step the sequential-auction protocol can produce, wage set to the poacher’s full value $q = p$). Both sit an order of magnitude above the data target of 0.109 (dashed) at every β_S ; the Bertrand floor declines only to about 0.98 at $\beta_S = 0.38$ and never reaches the target region. The step is set by the distance between successive match qualities, not by the wage rule, so no bargaining protocol repairs it.

The one mechanism that can fix it, and the tradeoff it forces. If the step is too large *because* redraws are global, the disciplined fix is to make them local: weight the redraw by a Gaussian kernel centred on the worker’s current match quality, with bandwidth h , so that a mover lands near where he already was and the step shrinks continuously as $h \rightarrow 0$. This is a single new parameter, it nests the memoryless model as $h \rightarrow \infty$, and it works for its intended purpose — the bandwidth tunes the EE step smoothly onto the target, reaching 0.139 against 0.109 at the estimate (Figure 2b). But the same locality that shrinks the step has a second, unavoidable consequence. Cross-sectional wage dispersion is a *squared* spread of match qualities around their mean; concentrating every redraw near the incumbent match pulls the equilibrium quality distribution inward, and the wage variance collapses with it. The two objects are driven by the *same* knob in opposite directions: the localisation that is needed to make the step small is exactly what destroys the dispersion. Estimated on the full moment set, the kernel model can reach the EE step only by sacrificing the rest — it attains $Q = 4.98$, missing both wage variances, both market tightnesses, and the wage-skewness moments by 60 to 100% (Figure 3). The canonical δ model, which keeps the memoryless redraw, fits every one of those dispersion, tightness, and skewness moments far better and attains $Q < 1$; the price it pays is the single overshoot on `ee_step_S`. The comparison is therefore not “a bad fit versus a good one” but a genuine modelling tension: one mechanism buys the wage step at the cost of wage inequality, the other buys wage inequality at the cost of the wage step, and no setting of the kernel buys both.

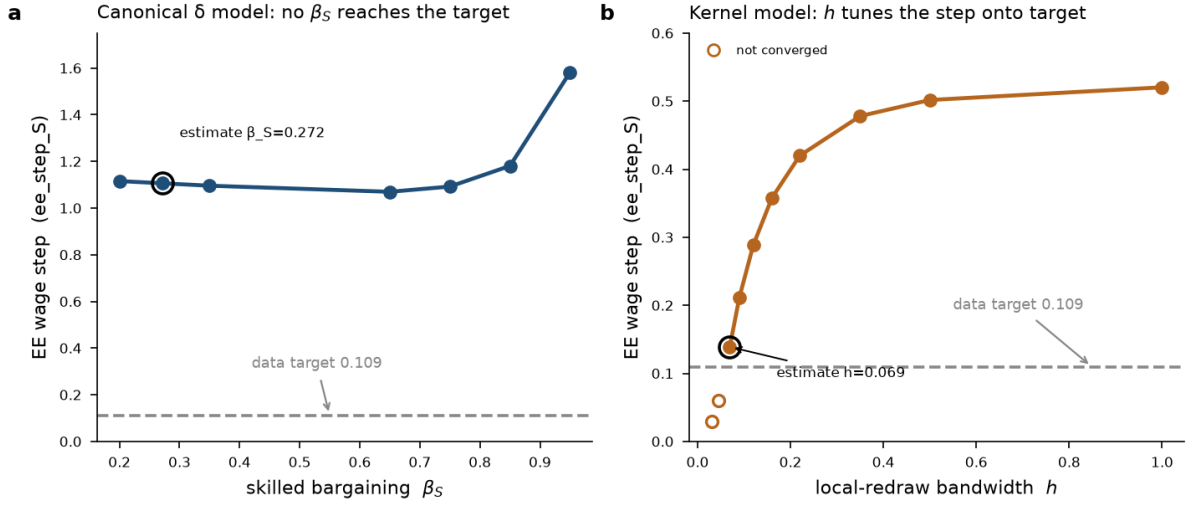


Figure 2: The skilled EE wage step against its two candidate levers, both evaluated at the base-covid estimates. **(a)** In the canonical memoryless-redraw model ($\delta = 0.869$), the step is a U-shaped function of the bargaining share β_S with a floor near 1.07 — roughly ten times the data target of 0.109 (dashed). No value of β_S brings the step near the target: bargaining is the wrong lever, because the step is set by how far a mover’s new match lies from his old one, not by how a given match’s surplus is divided. **(b)** In the local-kernel model, the bandwidth h tunes the step continuously; shrinking h localises the redraw so a mover lands *near* his current match, and the step falls through the target region, reaching 0.139 at the estimate $h = 0.069$ (open points below $h \approx 0.05$ did not converge).

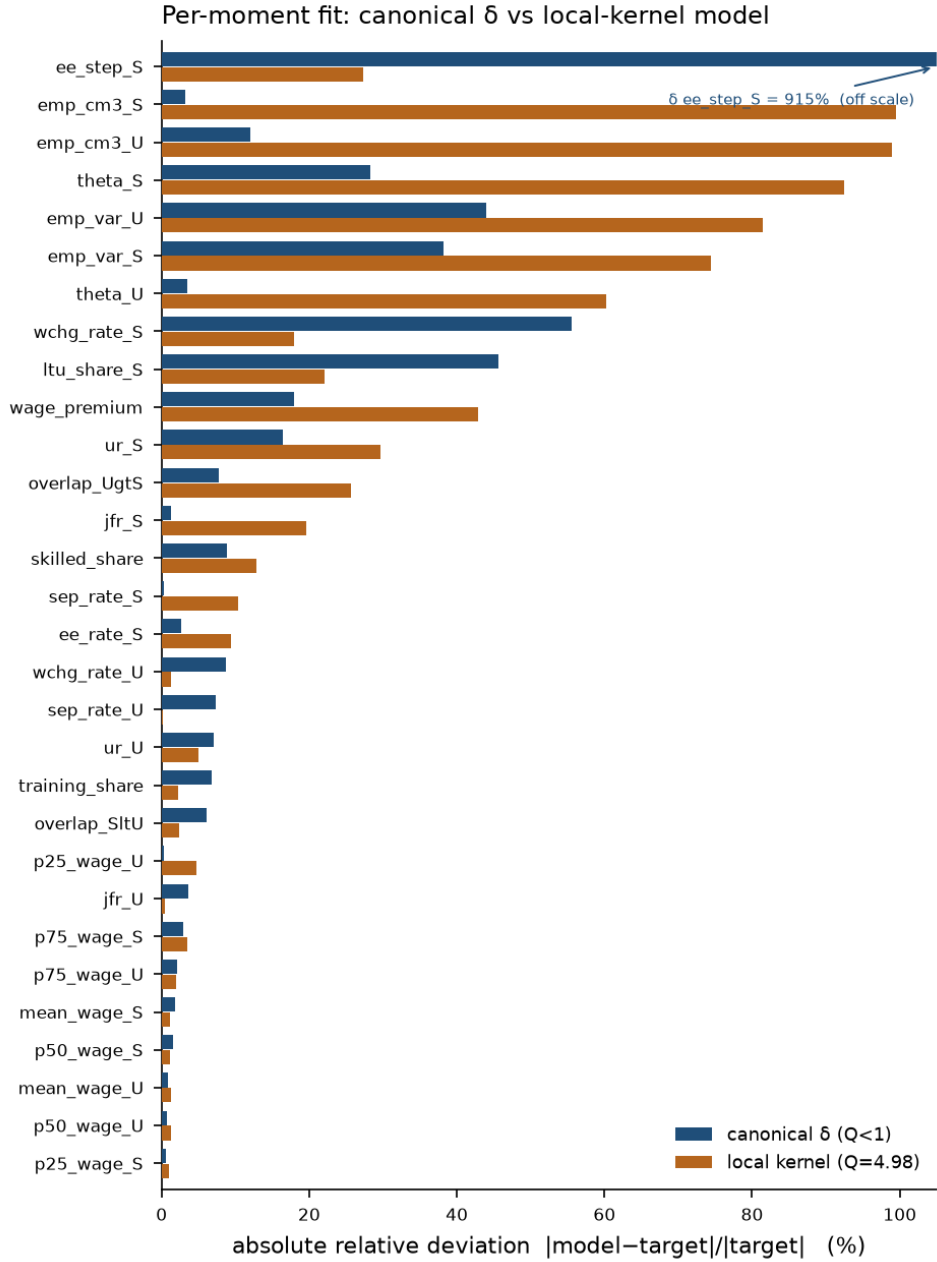


Figure 3: Per-moment fit of the two models against the 30 base-covid targets: absolute relative deviation $|m(\theta) - \hat{m}|/|\hat{m}|$ by moment, canonical δ model ($Q < 1$, β_S pinned) versus local-kernel model ($Q = 4.98$, β_S free). The kernel matches its one target moment **ee_step_S** but collapses the dispersion battery: skewness (**emp_cm3**), both tightnesses (**theta**), and both wage variances (**emp_var**) are missed by 60–100% as the localised redraw concentrates the equilibrium quality distribution. The δ model fits that whole block substantially better; its one large overshoot is **ee_step_S** (off scale, the held-out moment), the wage step the memoryless ladder mechanically cannot make small.

The choice, and why it is the defensible one. Faced with this tension there are three options. *Keep the kernel* and accept the collapse of wage dispersion: rejected, because cross-sectional wage inequality is a first-order feature of the labour-market data this model exists to speak to, and a fit that reproduces one job-ladder flow while flattening the wage distribution has traded a central object for a peripheral one. *Free the measurement-error scale* to prop the variance back up while the kernel handles the step: rejected, because that only launders the model’s missing structural dispersion into pure noise (it leaves the wage *skewness* untouched and drives the implied signal-reliability of wages far below any external estimate; Section 12). *Keep the memoryless redraw, target the dispersion the data identify, and drop ee_step_S out of the objective*: the choice made here. It is defensible on three grounds. First, the overshoot is understood — the redraw ladder mechanically cannot produce a small rung, so the moment is reported, not hidden, and the paper states plainly why the model overshoots it. Second, the model’s contribution is the mechanism and the estimation method, for which the wage *step* is a peripheral flow while the dispersion, tightness, and composition moments it does match are central. Reverting to the memoryless redraw is thus the outcome of pricing the experiment correctly: the kernel was the right thing to try, its tradeoff is real, and on the moments this data identifies the memoryless model dominates.

The measurement-error scale, and where it comes from. The defense above invokes the measurement-error scale $\sigma_{w,j}$, an object the earlier single-ability version of the model did not carry, so it is worth introducing before the discussion turns on it. It is the standard deviation of a classical, mean-zero noise term added to observed log wages, $\log w_j^{\text{obs}} = \log w_j + \sigma_{w,j} \varepsilon$ with $\varepsilon \sim N(0,1)$ (formalised in Section 12). Its role is to reconcile a structural fact of this class of model with the data: a frictional search model generates wage dispersion only through the equilibrium distribution of match quality, and that distribution is tighter than the empirical wage distribution (Hornstein et al., 2011), so without a noise term the model is forced to explain, structurally, more cross-sectional variance than it can produce. The scale $\sigma_{w,j}$ absorbs the residual dispersion the model cannot generate, so that the wage-variance moment disciplines the structural parameters without penalising the model for a gap that is intrinsic to the class.

The precedent, and my point of departure. Lise et al. (2016) introduce the same object for the same reason: they note that survey wages carry variation — reporting noise — beyond the economic sources their model produces (firm type, outside offers, layoffs, productivity shocks), and that leaving it unmodelled would bias the estimated structural parameters. They specify the noise as log-normal, common within a survey year and independent across years, and *estimate* its variance jointly with the rest of the model. They can do so because their moment set identifies the noise separately from the structural dispersion: as they put it, they “identify the variance of measurement error in wages from the variance of wages that the economic model is unable to reproduce,” using the profile of wage growth within and between jobs. My departure is that I *calibrate* $\sigma_{w,j}$ from outside the estimation rather than fit it, for an identification reason specific to my model and moment set, developed next.

Why the measurement-error scale and the matching elasticities are pinned, while the bargaining shares are estimated. The specification fixes the two matching elasticities and the two measurement-error scales (alongside the calibrated r, ν, ϕ), and *estimates* the two bargaining shares. The reasons differ across the two fixed groups, and neither is mere convenience.

The bargaining shares and the measurement scales are two ends of one identification problem: on a Nash-bargained cross-section both widen the observed wage distribution — a higher β_j passes more match-quality dispersion into wages, and a higher $\sigma_{w,j}$ adds noise — and the wage-variance moment cannot tell them apart, since $\text{Var}[\log w^{\text{obs}}] = \sigma_{\log}^2(\beta) + \sigma_w^2$. Exactly one of each pair must be fixed. Lise et al. (2016) face the same degeneracy in principle but resolve it by *identification*, not by pinning: their bargaining parameter is “primarily identified by wages following a job change, because it regulates the extent of a job-to-job wage increase,” and σ_w is then the residual variance the model cannot reproduce — two distinct footprints, so both can be estimated. That route is not open here, because the one moment that would identify β_S the way it identifies LMR’s β — the skilled EE wage step — is held out of the objective (the memoryless ladder overshoots it by an order of magnitude, above). With that footprint removed, β_j and $\sigma_{w,j}$ act on the same wage-variance moment and are not separately identified. I therefore fix $\sigma_{w,j}$ from outside (next paragraph), and this is precisely what frees the bargaining shares to be estimated on the wage moments rather than confounded with noise; both come in interior (Section 14). The matching elasticities are fixed for an unrelated reason given below.

First, the measurement-error scale σ_w . Lise et al. (2016), like much of the structural-search literature, absorbs the excess data variance into classical measurement error, and this model uses the same device: observed log wages are the model wage plus independent noise of variance σ_w^2 . The temptation is to *estimate* σ_w on the variance moment. I do not, because freeing it hands the model a degree of freedom I specifically want to withhold. The structural side must still generate *some* wage dispersion — the means, percentiles, the premium, and the market tightnesses all read the equilibrium quality distribution, so the structural parameters are never entirely free of the wage data. But a free σ_w would absorb whatever gap remained on the variance moment *itself*, so that this one moment no longer disciplined the structural parameters directly; the wage variance would stop pulling on β_j and the quality-distribution shape and instead be split off into noise. That defeats the purpose of targeting it. I want the wage variance to keep disciplining the structural side, only softened, so I pin σ_w from outside the estimation rather than free it. It is set from a wage-signal reliability λ_w through $\sigma_w^2 = (1 - \lambda_w) \widehat{\text{Var}}[\log w^{\text{obs}}]$, with λ_w taken from the validation literature and set at $\lambda_w = 0.78$ (Section 12 states the range this comes from). This scale is genuinely load-bearing rather than cosmetic: with $\sigma_w = 0$ the model must generate the observed wage variances structurally, and reproducing the unskilled variance alone then requires a bargaining share $\beta_U \approx 0.75\text{--}0.8$ — outside any defensible range — whereas the calibrated specification leaves β_U interior at about 0.54. The measurement-error scale is precisely what keeps the estimated bargaining shares economically sensible. With σ_w fixed, the variance moment still informs the structural parameters without penalising the model for a gap it cannot structurally close (developed in Section 12).

Second, the bargaining shares β_U and β_S — these are estimated, not pinned. The bargaining share is itself a dispersion lever, and in earlier specifications — before the measurement scale was pinned and before the within-job wage-change moments entered — a free β railed toward its bound, chasing unreachable variance. Two changes removed that pathology. First, fixing σ_w from outside (below) settles the β – σ_w degeneracy on the variance moment, so the wage moments now pull on β rather than on the noise scale. Second, the within-job wage-change hazard `wchg_rate.S` pins the skilled shock rate λ_S directly, breaking the β_S – λ_S co-movement that had produced the corner. With both in place I estimate both shares, and both land interior: $\beta_S \approx 0.32\text{--}0.44$ and $\beta_U \approx 0.19\text{--}0.54$ across the base windows, neither at a bound. Identification runs through market tightness and the wage distribution rather than through the EE wage step

(which the memoryless ladder overshoots and which stays out of the objective): β_U 's strongest Jacobian entry is unskilled tightness `theta.U`, and β_S loads on the skilled dispersion block. That the freely-estimated β_U sits near the Lise et al. (2016) lower-education estimate (0.188), obtained from different data and a different moment set, is corroboration of the estimate rather than a value imposed on it.

Third, the matching elasticities η_U and η_S . Separating a matching elasticity from the matching efficiency μ_j requires variation in market tightness — movement along the matching function — but the estimation targets only within-window cross-sectional moments, which pin the *level* of job-finding and tightness, not the *slope* of the matching function. No moment in the set identifies η_j , so I fix both at $\eta_U = \eta_S = 0.5$, the conventional value for an unidentified elasticity and within the empirical range surveyed by Petrongolo and Pissarides (2001). It is the same move Lise et al. (2016) make and for the same reason: they note the matching function “can only be identified as a result of fluctuations in the aggregate vacancies and unemployment and consequently cannot be estimated” in an environment without aggregate fluctuations, and so take it from the literature at an elasticity of one-half (imposed by their symmetric square-root form $M(u, v) = \alpha\sqrt{uv}$, which carries no free elasticity parameter). I adopt the practice — assuming away an object the moment set cannot identify — rather than importing a value as a structural claim. Taken together, the fixed parameters are not choices withheld from the optimiser but parameters the moment set cannot identify (the elasticities) and one scale deliberately calibrated from outside so that a moment the model cannot fully match still informs the estimation without dominating it (the measurement error). Fixing that scale is what lets the bargaining shares be estimated rather than pinned: with the noise absorbed externally, the wage moments identify β_U, β_S , and both come in interior. The spec is, in this sense, as tight as the data and the model together allow.

2 Global environment

Time is continuous. All agents discount at rate $r > 0$.

2.1 Aggregate scale and sectoral productivity shifters

A single aggregate productivity scale $A > 0$ multiplies output in both sectors; it sets the overall dollar level of production and wages and carries no cross-sectional shape information. Net of A , there are two sector-specific productivity shifters $P_U > 0$ and $P_S > 0$. Both are constant scalars. The two sectors differ only in *which ability they read*: the unskilled technology is linear in the unskilled aptitude a_U , and the skilled technology is linear in the skilled aptitude a_S ,

$$\pi_U(a_U, p) = A P_U a_U p, \quad \pi_S(a_S, p) = A P_S a_S p, \quad (a_U, a_S) \in [0, 1]^2, \quad p \in (0, 1]. \quad (1)$$

Both loadings are linear and strictly increasing in own ability: there is no curvature parameter, no floor, and no convexity. Those devices existed in the single-ability model only to manufacture separation, and ρ_x now supplies separation directly (Proposition 5). Consequently output is strictly increasing in own productivity in both markets — the “decreasing in productivity” pathology is impossible by construction.

The overall dollar scale is carried by A (which also scales the institutional flow values and vacancy costs); P_U and P_S move one sector each, so relative sectoral productivity is identified from cross-sector level comparisons. Both shifters are parameterised in logs in estimation, mirroring

the log scale A and the training-cost coefficient c : the stored parameters are $\log P_U$ and $\log P_S$, so the optimiser searches over $\mathbb{R}$ while positivity is automatic. All equations below are written in the levels P_U, P_S .

2.2 Demographic turnover

Workers exit the economy at Poisson rate $\nu > 0$ and are replaced one-for-one by newborn untrained workers drawn from the population type distribution $\ell(a_U, a_S)$ (Section 4). This delivers a stationary population composition. The rate ν is the model's only always-on renewal clock; it fires identically in every stationary equilibrium and along every transition path.

2.3 Stationarity

The model admits a stationary equilibrium in which the cross-sectional distribution of workers across types and states is time-invariant. The transition-dynamics section (Section 15) studies the path generated by an unexpected, permanent change in some parameters at a date T .

3 Parameter classification

Parameters fall into four groups, following the multi-regime estimation strategy. The classification identifies which parameters are externally set, which are estimated once and held fixed across windows, and which are re-estimated within each window. All parameters appear without a regime argument in the equations below.

3.1 Calibrated / fixed

Symbol	Name	Source / calibration
r	Discount rate	5% annual real rate (risk-free rate)
η_U, η_S	Matching elasticities	Fixed at 0.5 (Petrongolo and Pissarides, 2001)

3.2 Pre-estimated from data

Unless stated otherwise, every quantity is measured separately on each of the four windows.

Symbol	Name	Source / method
ϕ	Training completion rate	NSC/IPEDS enrollment + NCES median time-to-degree, enrollment-weighted across 4-year and 2-year programmes. One value, pooled over all Fall semesters, used in all four windows
ν	Demographic turnover rate	CPS Basic Monthly life table (16–64): inverse weighted-mean remaining working life to age 65. Estimated once per baseline window; each crisis window takes its own baseline's value
$\sigma_{w,U}, \sigma_{w,S}$	Wage measurement error	Calibrated from the log-wage residual variance to break the β – σ_w degeneracy (Section 12)

The training completion rate is the enrolment-weighted mean of the per-track completion rates,

$$\hat{\phi} = \frac{E_{4\text{yr}}/d_{4\text{yr}} + E_{2\text{yr}}/d_{2\text{yr}}}{E_{4\text{yr}} + E_{2\text{yr}}},$$

where $E_{4\text{yr}}$ and $E_{2\text{yr}}$ are US total Fall enrolments at all four-year and all two-year institutions from the NSC/IPEDS enrolment universe, averaged over all available Fall semesters, and $d_{4\text{yr}} = 49$, $d_{2\text{yr}} = 37$ months are the NCES median times to a bachelor's and an associate's degree. A single $\hat{\phi}$ is used in all four windows. The demographic turnover rate is a life table on CPS Basic Monthly labour-force participants aged 16–64: remaining working life is $\max(65 - a, 0)$ years at age a , and $\hat{\nu} = 1/\bar{R}$ with $\bar{R}$ the weight-averaged remaining life in months. It is estimated once on each baseline window, and each crisis window takes the $\hat{\nu}$ of its own baseline.

3.3 Deep structural parameters

Symbol	Name	Interpretation
a_ℓ, b_ℓ	Ability marginal (Beta)	Worker heterogeneity (per-axis shape)
ρ_x	Skill-copula correlation	Interdependence of the two abilities
$\tilde{b}_U$	Unskilled outside flow (rel. to A)	Institutional UBI and UI policy
$\tilde{b}_T$	Training flow (rel. to A)	Training stipend / support
$\tilde{b}_S$	Skilled outside flow (rel. to A)	Institutional UBI and UI policy

These govern the *fundamentals* assumed invariant across crisis episodes. The new deep parameter is ρ_x : the correlation of the Gaussian skill copula (Section 4). It is a population characteristic — how aligned unskilled and skilled aptitudes are across people — and is therefore held fixed across windows. In the symmetric baseline the two abilities share the single marginal pair (a_ℓ, b_ℓ) ; an asymmetric fallback (Section 4) would replace this with $(a_{\ell U}, b_{\ell U}, a_{\ell S}, b_{\ell S})$.

3.4 Regime-specific parameters (re-estimated within each window)

Symbol	Name	Interpretation
c	Training cost coefficient	Cost of education
A	Aggregate production scale	Overall TFP / demand level
P_U, P_S	Productivity shifters	Relative sectoral productivity
$\tilde{k}_U, \tilde{k}_S$	Vacancy costs (rel. to A)	Hiring conditions
λ_U, λ_S	Shock arrival rates	Productivity re-draw rate
ξ_U, ξ_S	Exogenous separation hazards	Baseline layoff rate
α_U	Unskilled damage shape	Damage distribution
a_Γ, b_Γ	Skilled quality shape	Shared shape of offer and shock distributions
δ	Shock-support scale	Offer/shock gap; $\delta \rightarrow 1$ collapses to a single distribution
μ_U, μ_S	Matching efficiencies	Matching-function level
β_U, β_S	Bargaining weights	Wage-setting institutions
σ_S	OJS flow cost	On-the-job search cost

These capture the *aggregate state* (A, P_U, P_S), the *matching and wage-setting technology*, and the *training technology*. Relative to the single-ability notes, the skilled-loading curvature

has been removed from this block (production is now linear in a_S), and no per-window shape parameter replaces it; the separation of the markets is handled by the deep parameter ρ_x .

4 Workers, firms, and types

4.1 Two abilities and the joint distribution

Each worker is characterised by an innate, permanent pair of abilities

$$\mathbf{a} = (a_U, a_S) \in \mathcal{A} = [0, 1]^2,$$

fixed over time, where a_U is aptitude for unskilled work and a_S is aptitude for skilled work. Abilities do not change with employment, training, or age; training changes *market access only*, not $\mathbf{a}$.

Marginals. Each ability has a Beta marginal. In the baseline *symmetric* specification the two marginals share parameters,

$$a_U \sim \text{Beta}(a_\ell, b_\ell), \quad a_S \sim \text{Beta}(a_\ell, b_\ell),$$

with common CDF F_ℓ and density f_ℓ . As a fallback, invoked only if the symmetric version cannot fit the cross-market overlaps, the two axes take their own shapes $(a_{\ell U}, b_{\ell U})$, $(a_{\ell S}, b_{\ell S})$.

Copula. The joint distribution is a Gaussian copula with a single correlation $\rho_x \in (-1, 1)$ on the Gaussian ranks:

$$L(a_U, a_S) = \Phi_{\rho_x}(\Phi^{-1}(F_\ell(a_U)), \Phi^{-1}(F_\ell(a_S))),$$

where Φ_{ρ_x} is the bivariate standard normal CDF with correlation ρ_x . The joint density is

$$\ell(a_U, a_S) = f_\ell(a_U) f_\ell(a_S) c_{\rho_x}(F_\ell(a_U), F_\ell(a_S)), \quad c_{\rho_x}(u, v) = \frac{\exp\left(-\frac{\rho_x^2(\zeta_u^2 + \zeta_v^2) - 2\rho_x \zeta_u \zeta_v}{2(1 - \rho_x^2)}\right)}{\sqrt{1 - \rho_x^2}}, \quad (2)$$

with $\zeta_u = \Phi^{-1}(u)$, $\zeta_v = \Phi^{-1}(v)$. Newborns are drawn from $\ell(a_U, a_S)$.

Proposition 1 (Nesting of the single-ability model). *As $\rho_x \rightarrow 1$ the copula concentrates on the diagonal $a_U = a_S$; the two abilities collapse to one index $x \equiv a_U = a_S$ and RoySearch reduces to the previous one-dimensional model (with the linear production of (1)). The single-ability model is thus the $\rho_x = 1$ special case, giving a clean internal robustness check.*

4.2 Firms

There are two labour markets, U and S . Both markets have homogeneous firms. A firm in market j posts a vacancy; on contact with a worker it reads only that market's ability (a_U in U , a_S in S) and match quality p .

4.3 Market access, training, and cross-market search

Market access via training. Workers participate in either the unskilled market U (untrained status) or the skilled market S (trained status). Training changes market access only.

Training decision — a frontier, not a cutoff. Training can be initiated only by untrained workers in unskilled unemployment. Because the payoff to training is driven by the skilled aptitude a_S while the value of remaining unskilled is driven by a_U , the optimal policy is characterised by a *frontier curve* in the (a_U, a_S) plane rather than a scalar cutoff:

$$\tau(a_U, a_S) \in \{0, 1\}, \quad \tau(a_U, a_S) = 1 \iff -c(a_S) + T(a_S) \geq U_U^{\text{search}}(a_U),$$

with $\tau = 1$ (train) *above* the frontier (high a_S , low a_U) and $\tau = 0$ (stay unskilled) *below* it. The frontier $\mathcal{F} = \{(a_U, a_S) : -c(a_S) + T(a_S) = U_U^{\text{search}}(a_U)\}$ replaces the old cutoff $\bar{x}$; its slope is characterised in Proposition 6.

Cross-market search by the skilled unemployed — always priced. A trained worker in skilled unemployment always has, and always prices, the option of directing search to the unskilled market using a_U :

$$d(a_U, a_S) \in \{0, 1\}, \quad d(a_U, a_S) = 1 \iff \text{the worker searches the unskilled market.}$$

Searching is exclusive: at each instant the worker is contacted in one market only, at rate $\kappa_S = \theta_S q_S(\theta_S)$ if $d = 0$ or $f_U = \theta_U q_U(\theta_U)$ if $d = 1$. The optimal d is a comparison of two candidate unemployment values (Section 9). This is a *standard directed-search primitive*, present identically in steady state and in transition; nothing is switched on or off. Its steady-state behaviour and the elimination of the stipend-farming pathology are treated in Section 16. Skilled-employed workers do not search the unskilled market.

Loss of training upon cross-market matching. A skilled-unemployed worker who matches in the unskilled market is treated as *untrained* from the moment of acceptance: the bargained wage uses U_U as the outside option, the firm receives the standard $J_U(a_U, 1)$, and upon a later separation the worker re-enters untrained unemployment and may re-initiate training (paying $c(a_S)$) if her type warrants it. We call this the “training is obsolete” interpretation of the cross-market move.

5 Training technology

Eligibility. Only untrained workers in unskilled unemployment may initiate training.

Cost. Initiating training requires a one-time flow cost that depends on the *skilled* aptitude:

$$c(a_S) = c(1 - a_S) e^{-a_S}, \quad c'(a_S) < 0. \quad (3)$$

Workers who are more apt at skilled work train more cheaply — the economically appropriate argument is aptitude for the skill being acquired, a_S , not the unskilled ability. The coefficient $c > 0$ is a *linear level multiplier* on the fixed shape $(1 - a_S) e^{-a_S}$; a higher c raises the cost at every a_S and $c = 0$ removes the cost entirely. Like the aggregate scale A , c is a level that is estimated *on the log scale* — the free parameter and the codebase carry $\tilde{c} = \ln c$ (hence the $\tilde{c}$ notation), so that positivity is automatic and the parameter is comparable to A ; the two log-scale parameters of the model are A and c .

The cost is a *psychological*, fixed-utility cost of effort, *not* a pecuniary outlay: it does *not* carry the aggregate scale A , unlike the flow values b_j , the vacancy costs k_j , and sectoral output, all of

which scale with A (Section 6). Consequently the training margin $-c(a_S) + T(a_S) \geq U_U^{\text{search}}(a_U)$ is *not* homogeneous in A : the value of training T and the outside option U_U^{search} grow with the aggregate scale while the cost does not, so the margin shifts with the wage level rather than being scale-invariant. The estimated c places the cost on a scale comparable to those terms at the estimated output level, so it does not drop out of the frontier, and the frontier is governed by comparative advantage, foregone income, and this scale-shifting cost.

Completion. Training completes at Poisson rate $\phi > 0$; on completion the worker becomes trained and enters skilled unemployment.

Flow payoffs. b_U , b_T , b_S denote flow payoffs while unskilled unemployed, in training, and skilled unemployed. A skilled-unemployed worker receives b_S regardless of the search-market choice d .

6 Production technologies

Output is given by (1):

$$\pi_U(a_U, p) = A P_U a_U p, \quad \pi_S(a_S, p) = A P_S a_S p.$$

The pure-Roy mapping — unskilled output reads only a_U , skilled output reads only a_S — is the source of the model’s tractability: unskilled value objects are functions of a_U alone, and the *employed* skilled value objects are functions of a_S alone (Sections 7–9). The two-dimensional type enters only at the two decision boundaries (the training frontier τ and the cross-market choice d , each comparing across abilities), in the value of skilled unemployment $U_S(a_U, a_S) = \max\{U_S^{(0)}(a_S), U_S^{(1)}(a_U)\}$ where the abandon-skill option reads a_U (Section 9), and in aggregation against the joint density (2).

Both production functions carry the common scale A . Institutional flow values are expressed relative to it, $b_j = \tilde{b}_j A$ for $j \in \{U, T, S\}$ with the estimated object the dimensionless $\tilde{b}_j$. The vacancy posting costs use the Lise et al. (2016) convention: they are denominated in *months of average sectoral output*, $k_j \bar{\pi}_j$ for $j \in \{U, S\}$, where k_j is the dimensionless estimated cost and $\bar{\pi}_j \equiv \mathbb{E}[\pi_j]$ is mean sectoral flow output,

$$\bar{\pi}_U = A P_U \mathbb{E}_\ell[a_U] \mathbb{E}_G[p], \quad \bar{\pi}_S = A P_S \mathbb{E}_\ell[a_S] \mathbb{E}_\Gamma[p].$$

Thus the dollar cost of holding a vacancy open for a month equals k_j times what an average filled job in that sector produces in a month; k_j is a pure ratio of two flows and so is scale-free (the dollar cost inherits the aggregate scale, sectoral productivity, and the ability/quality means through $\bar{\pi}_j$, while k_j itself does not move with them).

7 Unskilled labour market

Under pure-Roy, every unskilled-market object is a function of a_U ; the two-dimensional type enters only through the frontier τ (which determines who is untrained) and through the cross-market inflow du_S .

7.1 Matching

Let $\mathcal{V}_U$ be the mass of unskilled vacancies. The relevant seeker mass is the augmented pool

$$\tilde{\mathcal{U}}_U \equiv \iint (u_U(a_U, a_S) + d(a_U, a_S) u_S(a_U, a_S)) da_U da_S,$$

which adds the skilled-unemployed who have chosen $d = 1$. Tightness is $\theta_U = \mathcal{V}_U / \tilde{\mathcal{U}}_U$. Matching is Cobb–Douglas with $\mu_U > 0$, $\eta_U \in (0, 1)$:

$$q_U(\theta_U) = \mu_U \theta_U^{-\eta_U}, \quad f_U \equiv \theta_U q_U(\theta_U) = \mu_U \theta_U^{1-\eta_U}.$$

A contacted worker has unskilled ability a_U with probability proportional to the augmented seeker density at a_U ; the firm's filled-job value depends only on a_U .

7.2 Match productivity and quality shocks

New matches start at frontier efficiency $p_0 = 1$. Quality shocks arrive at Poisson rate λ_U ; on a shock the match quality is *redrawn*, with the new quality p' independent of the current one (a memoryless redraw),

$$p' \sim \text{Beta}(\alpha_U, 1), \quad G(p) = p^{\alpha_U}, \quad p \in [0, 1],$$

equivalently $p' = e^{-\chi}$ with $\chi \sim \text{Exp}(\alpha_U)$. Because a new match is born at the top of the support, the first shock can only (weakly) lower quality; from an interior match, by contrast, the redraw is memoryless and may raise or lower p , reverting toward the mean draw $\mathbb{E}_G[p] = \alpha_U / (\alpha_U + 1)$. A match is endogenously destroyed whenever the redraw falls below the reservation, $p' < p^*(a_U)$, and is destroyed exogenously at the quality-independent rate ξ_U . The unskilled market has no on-the-job search, so the only route back to $p = 1$ is separation into unemployment and a fresh match.

7.3 Value functions and the training decision

Let U_U be the value of untrained unskilled unemployment, U_U^{search} the value of unskilled search, T the value of training, $E_U(a_U, p)$ the worker value of employment, $J_U(a_U, p)$ the firm value, V_U the vacancy value. An untrained unemployed worker of type (a_U, a_S) chooses between unskilled search and training:

$$U_U(a_U, a_S) = \max\{U_U^{\text{search}}(a_U), U_U^{\text{train}}(a_S)\}, \quad U_U^{\text{train}}(a_S) := -c(a_S) + T(a_U, a_S). \quad (4)$$

The search value depends only on a_U . Training completes at rate ϕ into skilled unemployment, and the completing worker collects the value of skilled unemployment U_S — which, as (21) makes explicit, is itself the discrete-choice maximum $U_S(a_U, a_S) = \max\{U_S^{(0)}(a_S), U_S^{(1)}(a_U)\}$ over staying skilled and abandoning skill to cross to the unskilled market:

$$(r + \phi + \nu) T(a_U, a_S) = b_T + \phi U_S(a_U, a_S) \quad (5)$$

A worker in training collects the flow b_T ; at rate ϕ she completes and enters skilled unemployment, worth U_S ; at rate ν she exits. Nothing else happens in the training state — there is no search while training, which is why no meeting rate appears. Both options are priced into the training

decision. In equilibrium, however, the maximum resolves to the stay-skilled branch *on the entire training region*: the workers who choose to train are those with high a_S relative to a_U (Proposition 6), and there $U_S^{(0)}(a_S) \geq U_S^{(1)}(a_U)$, so the abandon-skill branch never binds for anyone who has just trained. The cross-back option $U_S^{(1)}(a_U)$ is carried but inactive, and the a_U -dependence of the training *decision* then enters solely through the unskilled search value $U_U^{\text{search}}(a_U)$. The cross-market branch becomes active only when a crisis moves the frontier in a direction that makes training relatively less attractive to searching the unskilled market. Thus, the skill abandoning mechanism is only relevant during a transition (Section 15). Unskilled search yields b_U and contacts vacancies at rate f_U :

$$(r + \nu) U_U^{\text{search}}(a_U) = b_U + f_U (E_U(a_U, 1) - U_U^{\text{search}}(a_U)). \quad (6)$$

Over the next instant an unskilled searcher either stays unemployed, collecting the flow b_U , or — with probability f_U — meets a vacancy and starts a match at frontier quality $p = 1$, gaining $E_U(a_U, 1) - U_U^{\text{search}}(a_U)$. The left-hand side discounts at r and accounts for demographic exit at ν , whose continuation value is zero. Define the net training gain

$$\Delta_T(a_U, a_S) := -c(a_S) + T(a_U, a_S) - U_U^{\text{search}}(a_U).$$

The training frontier is $\{\Delta_T = 0\}$ and $\tau(a_U, a_S) = \mathbf{1}\{\Delta_T(a_U, a_S) \geq 0\}$.

The two boundaries in steady state and in transition. The type space is partitioned by two boundaries, each the locus where a discrete-choice maximum switches branch: the training frontier τ of (4), separating training from unskilled search, and the abandonment frontier d of (21), separating staying skilled from crossing to the unskilled market. In a stationary equilibrium the two boundaries are aligned by self-selection: trainees are the high- a_S /low- a_U types, and on their region $U_S^{(0)}(a_S) \geq U_S^{(1)}(a_U)$, so $d = 0$ wherever $\tau = 1$. Each maximum therefore resolves to a single branch on the region its own population occupies. Below τ the unskilled-employed and unskilled-search populations have $U_U(a_U, a_S) = U_U^{\text{search}}(a_U)$, so every unskilled match object — wages, surplus, reservation quality — takes $U_U^{\text{search}}(a_U)$ as its outside option and is a function of a_U alone; above τ the skilled population has $U_S = U_S^{(0)}(a_S)$, so every skilled match object is a function of a_S alone. The full two-dimensional type enters the stationary equilibrium only through the location of the two boundaries, not through the value functions on either side. Out of steady state this alignment breaks: an unanticipated shift in the frontier places skilled workers on the $d = 1$ side, activating the abandonment branch and the cross-market flow it governs. The two-way coupling between the markets is thus a property of the transition, not of any stationary equilibrium.

7.4 Match values, bargaining, and endogenous separation

For $p \in (0, 1]$, worker and firm values satisfy, with base rate $r + \nu + \lambda_U + \xi_U$,

$$(r + \nu + \lambda_U + \xi_U) E_U(a_U, p) = w_U(a_U, p) + \xi_U U_U + \lambda_U \left[U_U G(p^*(a_U)) + \int_{p^*(a_U)}^1 E_U(a_U, p') dG(p') \right], \quad (7)$$

$$(r + \nu + \lambda_U + \xi_U) J_U(a_U, p) = A P_U a_U p - w_U(a_U, p) + \lambda_U \int_{p^*(a_U)}^1 J_U(a_U, p') dG(p'). \quad (8)$$

Each equation reads as an enumeration of what can happen to a match of quality p over the next instant. The left-hand side is the required return on the asset at the base rate $r + \nu + \lambda_U + \xi_U$: the match is discounted at r and terminates at the demographic rate ν , the shock rate λ_U , and the exogenous destruction rate ξ_U . On the right, for the worker: she collects the flow wage $w_U(a_U, p)$; with probability ξ_U the match is destroyed for reasons unrelated to its quality and she returns to unskilled unemployment, worth U_U ; with probability λ_U the quality is redrawn, and the redraw splits into the two outcomes inside the bracket — with probability $G(p^*(a_U))$ the new quality falls below the reservation and the match dissolves, again worth U_U , and otherwise she continues in a match of the new quality p' , worth $E_U(a_U, p')$. The firm's equation enumerates the same events with two differences: its flow is output net of the wage, $AP_U a_U p - w_U(a_U, p)$, and it has no continuation on either destruction branch — an exogenous destruction and a below-reservation redraw both extinguish the asset, which is why ξ_U appears on the base rate of J_U but nowhere on its right-hand side, and why the redraw integral runs only over the surviving region $[p^*(a_U), 1]$.

Wages solve Nash bargaining with worker weight $\beta_U \in (0, 1)$:

$$\beta_U J_U(a_U, p) = (1 - \beta_U)(E_U(a_U, p) - U_U). \quad (9)$$

A skilled-unemployed worker who matches in U is treated as untrained: her outside option is U_U and the firm receives the same $J_U(a_U, 1) = (1 - \beta_U)S_U(a_U, 1)$ as for a fresh untrained match, which lets unskilled free entry use a single firm-value surface. Endogenous separation is characterised by $p^*(a_U) \in (0, 1)$ with

$$J_U(a_U, p^*(a_U)) = 0, \quad J_U(a_U, p) \geq 0 \iff p \geq p^*(a_U). \quad (10)$$

7.5 Surplus, reservation rule, and wages

Define $S_U(a_U, p) \equiv (E_U(a_U, p) - U_U) + J_U(a_U, p)$. Adding (7)–(8) (the $\xi_U U_U$ term combines with ξ_U on the base rate to leave the outside-option flow $(r + \nu)U_U$ unchanged, so ξ_U enters only through the base rate):

$$(r + \nu + \lambda_U + \xi_U) S_U(a_U, p) = AP_U a_U p - (r + \nu) U_U + \lambda_U \int_{p^*(a_U)}^1 S_U(a_U, p') dG(p'). \quad (11)$$

Read term by term: the match generates output $AP_U a_U p$; it must cover the flow value of the worker's outside option, $(r + \nu)U_U$, which is what the pair forgoes by staying matched; and it carries the option value of a redraw, $\lambda_U \int_{p^*}^1 S_U dG$, since a shock is not purely a loss — it may deliver a better quality, and the pair keeps the draw only when it clears the reservation. The reservation rule sets $S_U(a_U, p^*) = 0$:

$$p^*(a_U) = \frac{(r + \nu) U_U - \lambda_U \int_{p^*}^1 S_U(a_U, p') dG(p')}{AP_U a_U}, \quad (12)$$

unchanged in form: no explicit ξ_U appears in the numerator. The exogenous hazard is nonetheless *not* neutral for the endogenous cutoff. It enters the surplus through the base rate $r + \nu + \lambda_U + \xi_U$, which discounts the continuation option value $\lambda_U \int_{p^*}^1 S_U dG$ more heavily; a smaller option value raises $p^*(a_U)$. So a higher ξ_U both adds a quality-independent layoff hazard *and* lifts the endogenous reservation, destroying marginal matches that were previously retained — the two channels reinforce rather than separate. This is the same coupling the skilled cutoff already

exhibits through ξ_S : the reservation formula carries no explicit ξ_j term, but the reservation still responds to ξ_j through the option value. What is genuinely additive is the *realised hazard* $\xi_U + \lambda_U G(p^*(a_U))$ conditional on the cutoff; ξ_U is not additive in its effect on p^* itself. The reservation is decreasing in a_U : a more able worker produces more at any quality, so a lower p still clears her outside option. The wage is

$$w_U(a_U, p) = \beta_U A P_U a_U p + (1 - \beta_U)(r + \nu) U_U^{\text{search}}(a_U), \quad (13)$$

linear in p ; skilled-detoured and original untrained workers are paid identically at the same (a_U, p) .

7.6 Vacancies, free entry, and tightness

Free entry ($V_U = 0$) gives

$$q_U(\theta_U) = \frac{k_U \bar{\pi}_U \tilde{U}_U}{\iint J_U(a_U, 1) (u_U(a_U, a_S) + d(a_U, a_S) u_S(a_U, a_S)) da_U da_S}, \quad (14)$$

which pins down θ_U given the augmented seeker composition. A higher A raises $J_U(a_U, 1)$ through the surplus and hence tightness.

7.7 Stationary cross-sectional distributions (untrained)

Densities are over the two-dimensional type. Let $u_U(a_U, a_S)$ be untrained unskilled unemployment, $t(a_U, a_S)$ the training density, and define the untrained-segment mass and employed density

$$m_U(a_U, a_S) \equiv \ell(a_U, a_S) - m_S(a_U, a_S), \quad e_U(a_U, a_S) := m_U(a_U, a_S) - u_U(a_U, a_S) - t(a_U, a_S),$$

with m_S from (17). Because match objects depend only on a_U , the flow rates (f_U , $\lambda_U G(p^*(a_U))$, τ) are constant along a_S within each region; the a_S -coordinate enters only through ℓ and the frontier.

Training.

$$0 = \tau(a_U, a_S) u_U(a_U, a_S) - (\phi + \nu) t(a_U, a_S) \implies t(a_U, a_S) = \frac{\tau(a_U, a_S)}{\phi + \nu} u_U(a_U, a_S). \quad (15)$$

Untrained unemployment.

$$(f_U + \tau(a_U, a_S) + \nu) u_U(a_U, a_S) = \nu \ell(a_U, a_S) + (\lambda_U G(p^*(a_U)) + \xi_U) e_U(a_U, a_S). \quad (16)$$

Both are inflow-equals-outflow statements for a cell of the type space. Trainees leave the training state either by completing, at ϕ , or by exiting, at ν , and are replenished by the untrained unemployed of that type who choose to train; the stationary training mass is therefore the training inflow spread over the mean time spent in the state. The untrained unemployed of a type leave by finding a job, at f_U , by entering training, at τ , or by exiting, at ν ; they arrive as newborns, at $\nu \ell$, or from a destroyed match. The separation hazard feeding that second inflow is the sum of the endogenous quality-shock component $\lambda_U G(p^*(a_U))$ — a redraw landing below reservation — and the exogenous baseline ξ_U . Separations of skilled-detoured workers enter automatically, since they are counted in e_U .

8 Unskilled-block equilibrium and algorithm

An *unskilled-block equilibrium* is a collection $\{\theta_U, U_U^{\text{search}}(\cdot), T(\cdot), U_U(\cdot, \cdot), E_U, J_U, p^*(\cdot), u_U, t\}$ satisfying:

- (E1) **Values and training.** T solves (5); U_U^{search} solves (6); $U_U = \max\{U_U^{\text{search}}(a_U), -c(a_S) + T(a_S)\}$ with $\tau = \mathbf{1}\{-c(a_S) + T(a_S) \geq U_U^{\text{search}}(a_U)\}$.
- (E2) **Match values and wages.** (7)–(8) and (9) (equivalently w_U from (13)).
- (E3) **Separation.** $p^*(a_U)$ solves (10).
- (E4) **Free entry.** θ_U solves (14).
- (E5) **Distributions.** (15)–(16) with $e_U = m_U - u_U - t$, $m_U = \ell - m_S$.

8.1 Solution algorithm (unskilled side)

The block is a two-layer fixed point. The inner loop solves the value functions given (θ_U, p^*) ; the outer loop updates (τ, p^*, θ_U) and the composition. The cross-market pool du_S is supplied by the skilled block and treated as exogenous within one unskilled solve. Because unskilled value objects are one-dimensional in a_U , the inner loop runs on the a_U -grid; the frontier $\tau(a_U, a_S)$ is evaluated by comparing the a_U -indexed search value to the a_S -indexed training value on the two-dimensional grid.

Step 1. Initialise. Guess $\theta_U^{(0)} > 0$, $p^{*,(0)}(a_U) \in (0, 1)$, $\tau^{(0)}$, $u_U^{(0)}$, $t^{(0)}$.

Step 2. Inner value iteration. Holding (θ_U, p^*) fixed, iterate on $(U_U^{\text{search}}, U_U, T, E_U, J_U)$: update $T(a_S)$ from (5) (taking U_S from the skilled block); form $U_U(a_U, a_S) = \max\{U_U^{\text{search}}(a_U), -c(a_S) + T(a_S)\}$ and τ ; solve (7)–(8) with (9) for $E_U(a_U, p)$, $J_U(a_U, p)$; update $U_U^{\text{search}}(a_U)$ from (6). Iterate to tolerance.

Step 3. Distributions. Update (u_U, t) from (15)–(16) on the two-dimensional grid, using the converged τ and current p^* .

Step 4. Reservation rule. Update $p^*(a_U)$ from (12); Anderson-accelerate.

Step 5. Tightness. Update θ_U from (14) with the carried du_S ; Anderson-accelerate.

Step 6. Convergence. Stop when $\max\{|\Delta\theta_U|, \|\Delta p^*\|_\infty, \|\Delta u_U\|_\infty\} < \text{tol}$; else return to Step 2.

Remark 1 (Two-block coupling). *Step 2 takes $U_S(\cdot)$ from the skilled block; the skilled block takes $m_S(\cdot)$, f_U , $E_U(\cdot, 1)$, and the pool du_S from the unskilled block. A global equilibrium iterates the two blocks until the link variables are consistent (Section 14).*

9 Skilled labour market

This section adapts the Pissarides (2000, Ch. 4) on-the-job-search framework to the segmented environment, with skilled objects indexed by the skilled aptitude a_S . The block features (i) idiosyncratic match quality p redrawn by shocks, (ii) endogenous separation via $p_S^*(a_S)$, (iii) on-the-job search (OJS) as a discrete choice with flow cost σ_S , and (iv) the cross-market search choice d for the skilled unemployed. Skilled separations arise from an exogenous hazard ξ_S and the endogenous mechanism.

9.1 Primitives

Trained population. Let $m_S(a_U, a_S)$ be the density of trained workers. In stationarity, incorporating the cross-market outflow,

$$m_S(a_U, a_S) = \frac{\phi t(a_U, a_S)}{\nu + d(a_U, a_S) f_U}. \quad (17)$$

For $d = 0$ this is $(\phi/\nu) t$; for $d = 1$ the extra outflow f_U enters the denominator. Exogenous separation ξ_S reshuffles mass within the segment and does not affect m_S .

Production. $\pi_S(a_S, p) = A P_S a_S p$ from (1): linear in a_S .

Two quality distributions: offer and shock. New skilled matches — whether formed out of unemployment or reached by a successful on-the-job search — draw their idiosyncratic quality from the *offer distribution*

$$\Gamma_o = \text{Beta}(a_\Gamma, b_\Gamma) \text{ on } [0, 1], \quad \text{density } \gamma_o, \text{ CDF } \Gamma_o.$$

When an existing match is hit by an idiosyncratic shock, its quality is redrawn from the *shock distribution*, which shares the offer shape but is compressed onto $[0, \delta]$ with $\delta \in (0, 1]$:

$$\Gamma_s(x) = \Gamma_o(x/\delta), \quad \gamma_s(x) = \frac{1}{\delta} \gamma_o(x/\delta), \quad \text{supp } \Gamma_s = [0, \delta]. \quad (18)$$

Equivalently $p_{\text{shock}} = \delta p_{\text{offer}}$: the shock draw is a rescaling of the offer draw, so the two distributions have identical shape and differ only in support (hence in location). The offer distribution stochastically dominates the shock distribution for every $\delta < 1$; the single-distribution skilled block is the $\delta \rightarrow 1$ case ($\Gamma_s = \Gamma_o$). The economics of this gap — why it is what activates the on-the-job-search ladder and endogenous separation — is developed in Section 11.

Shocks. Match quality is hit at rate λ_S ; the redraw is i.i.d. Γ_s . A match is destroyed if the redraw is below $p_S^*(a_S)$, which occurs with probability $\Gamma_s(p_S^*(a_S))$; or exogenously at rate ξ_S ; or by demographic exit ν .

OJS. An employed skilled worker may search at flow cost σ_S , receiving offers at rate $\kappa_S = \theta_S q_S(\theta_S)$, each a draw $\tilde{p} \sim \Gamma_o$, and quits iff $\tilde{p} > p$.

9.2 Matching and tightness

Let $u_S(a_U, a_S)$ be skilled unemployment and $e_S(a_U, a_S, p)$ employed density. Only $d = 0$ types search the skilled market when unemployed; employed skilled search via s^* . The skilled seeker mass is

$$\tilde{U}_S = \iint (1 - d) u_S da_U da_S + \iiint_0^1 s^*(a_U, a_S, p) e_S(a_U, a_S, p) dp da_U da_S,$$

$$\theta_S = \mathcal{V}_S / \tilde{U}_S, q_S(\theta_S) = \mu_S \theta_S^{-\eta_S}, \kappa_S = \theta_S q_S(\theta_S).$$

9.3 Two unemployment values and the directed-search choice

Let $U_S(a_S)$ be skilled unemployment; E_S^0, E_S^1 worker values without/with OJS; J_S^0, J_S^1 firm values; $E_S = \max\{E_S^0, E_S^1\}$, $J_S = J_S^*$. A skilled-unemployed worker either searches the skilled market, meeting a new match drawn from the *offer* distribution, or crosses to the unskilled market, accepting $E_U(a_U, 1) = U_U + \beta_U S_U(a_U, 1)$. The two options give two candidate unemployment values:

$$U_S^{(0)}(a_S) \equiv \frac{b_S + \kappa_S \beta_S \int_{p_S^*(a_S)}^1 S_S(a_S, \tilde{p}) d\Gamma_o(\tilde{p})}{r + \nu}, \quad (19)$$

$$U_S^{(1)}(a_U) \equiv \frac{b_S + f_U E_U(a_U, 1)}{r + \nu + f_U}, \quad (20)$$

under Nash bargaining ($E_S - U_S = \beta_S S_S$). Each branch is the value of a search strategy. Staying skilled, the worker collects b_S and at rate κ_S meets an offer, keeping her share β_S of the surplus on draws that clear reservation — the integral runs against the *offer* distribution Γ_o , since an unemployed worker who matches draws a fresh quality. Crossing to the unskilled market, she collects b_S while searching and at rate f_U starts an unskilled match at frontier quality, worth $E_U(a_U, 1)$; the form of $U_S^{(1)}$ embeds always accepting that offer, which is why f_U appears on its base rate. **Crucially, the two candidates read different abilities:** the skilled branch reads a_S , the unskilled branch reads a_U . The worker picks the higher:

$$d(a_U, a_S) = \mathbf{1}\{U_S^{(1)}(a_U) > U_S^{(0)}(a_S)\}, \quad U_S(a_U, a_S) = \max\{U_S^{(0)}(a_S), U_S^{(1)}(a_U)\}. \quad (21)$$

Remark 2 (The directed-search choice is always priced). *Equation (21) holds in every stationary equilibrium and along the transition alike; no option is added or removed between regimes. In the employed-worker HJBs below, U_S is understood as this discrete-choice max. Because employed skilled objects otherwise read only a_S , we write $U_S(a_S)$ in those equations, with the understanding that on the $d = 1$ region the max resolves to $U_S^{(1)}(a_U)$.*

9.4 Employed-worker HJBs

With base rate $r + \nu + \lambda_S + \xi_S$:

$$(r + \nu + \lambda_S + \xi_S) E_S^0(a_S, p) = w_S^0(a_S, p) + \xi_S U_S + \lambda_S \left[U_S \Gamma_s(p_S^*) + \int_{p_S^*}^{\delta} E_S(a_S, p') d\Gamma_s(p') \right], \quad (22)$$

$$(r + \nu + \lambda_S + \xi_S) E_S^1(a_S, p) = (w_S^1(a_S, p) - \sigma_S) + \xi_S U_S + \lambda_S \left[U_S \Gamma_s(p_S^*) + \int_{p_S^*}^{\delta} E_S(a_S, p') d\Gamma_s(p') \right] + \kappa_S \int_{\max\{p, p_S^*\}}^1 (E_S(a_S, \tilde{p}) - E_S^1(a_S, p)) d\Gamma_o(\tilde{p}), \quad (23)$$

$$(r + \nu + \lambda_S + \xi_S) J_S^0(a_S, p) = AP_S a_S p - w_S^0(a_S, p) + \lambda_S \int_{p_S^*}^{\delta} J_S(a_S, p') d\Gamma_s(p'). \quad (24)$$

$$(r + \nu + \lambda_S + \xi_S + \kappa_S(1 - \Gamma_o(p))) J_S^1(a_S, p) = AP_S a_S p - w_S^1(a_S, p) + \lambda_S \int_{p_S^*}^{\delta} J_S(a_S, p') d\Gamma_s(p'). \quad (25)$$

The four equations enumerate the events facing a skilled match, and differ only in which events are available. In E_S^0 the worker does not search on the job: she collects w_S^0 ; at rate ξ_S the match is destroyed regardless of quality and she enters skilled unemployment, worth U_S ; at rate λ_S her quality is redrawn from the shock distribution, and the bracket splits that redraw into dissolution when it lands below the reservation (probability $\Gamma_s(p_S^*)$, worth U_S) and continuation at the new quality otherwise. E_S^1 adds the two consequences of searching on the job: the flow cost σ_S is deducted, and at rate κ_S she meets an outside offer, moving whenever the offer beats her current position — the last integral, which runs over offers above $\max\{p, p_S^*\}$ and pays the gain $E_S - E_S^1$. The firm equations mirror these events without the worker's outside option: the flow is output net of the wage, and both destruction routes extinguish the asset, so only the surviving part of the redraw appears. The one extra term in J_S^1 is on the left: the firm of an on-the-job searcher loses the match whenever she draws a better offer, at rate $\kappa_S(1 - \Gamma_o(p))$, which enters as an additional termination hazard on its base rate. The cross-market mechanism enters solely through U_S in (21).

9.5 Policies, surplus, and wages

Overall surplus $S_S(a_S, p) = \max\{S_S^0, S_S^1\}$. Separation: $S_S(a_S, p_S^*(a_S)) = 0$, $S_S \geq 0 \iff p \geq p_S^*(a_S)$. OJS: $s^*(a_S, p) = \mathbf{1}\{E_S^1 \geq E_S^0\}$ with cutoff $p_S^{\text{oj}}(a_S) \geq p_S^*(a_S)$. The surplus equations are

$$(r + \nu + \lambda_S + \xi_S) S_S^0(a_S, p) = AP_S a_S p - (r + \nu) U_S + \lambda_S \int_{p_S^*(a_S)}^{\delta} S_S(a_S, \tilde{p}) d\Gamma_s(\tilde{p}), \quad (26)$$

$$(r + \nu + \lambda_S + \xi_S + \kappa_S(1 - \Gamma_o(p))) S_S^1(a_S, p) = AP_S a_S p - (r + \nu) U_S - \sigma_S + \lambda_S \int_{p_S^*(a_S)}^{\delta} S_S(a_S, \tilde{p}) d\Gamma_s(\tilde{p}) + \kappa_S \beta_S \int_p^1 S_S(a_S, p') d\Gamma_o(p'), \quad (27)$$

The no-search surplus contains output, minus the flow value of the outside option the pair forgoes, plus the option value of a shock redraw. The searching surplus subtracts the search cost σ_S and adds the expected gain from moving up the ladder, and carries the extra poaching hazard on its base rate: the pair loses the match whenever a better offer arrives. The difference between the two is what the on-the-job-search cutoff $p_S^{\text{oj}}(a_S)$ balances — searching is worth its cost only from a low enough current quality.

and the wage schedules

$$w_S^0(a_S, p) = \beta_S AP_S a_S p + (1 - \beta_S)(r + \nu) U_S, \quad (28)$$

$$w_S^1(a_S, p) = \beta_S AP_S a_S p + (1 - \beta_S)(r + \nu) U_S + (1 - \beta_S) \sigma_S - \beta_S(1 - \beta_S) \kappa_S \int_p^1 S_S(a_S, p') d\Gamma_o(p'). \quad (29)$$

9.6 Vacancies and free entry

With skilled seeker density $\tilde{u}_S(a_U, a_S) = (1 - d)u_S + \int_0^1 s^* e_S dp$ and expected firm surplus $\mathcal{J}_S$ (the a_S -analogue of the single-ability expression, integrating $\int_{p_S^*}^1 \mathcal{J}_S d\Gamma_o$ over the seeker density — a new hire draws its quality from the offer distribution), free entry is

$$q_S(\theta_S) \mathcal{J}_S = k_S \bar{\pi}_S. \quad (30)$$

9.7 Stationary distributions (trained)

$m_S(a_U, a_S) = u_S(a_U, a_S) + \int_{p_S^*(a_S)}^1 e_S(a_U, a_S, p) dp$. With endogenous destruction hazard $\delta_S^{\text{end}}(a_S) = \lambda_S \Gamma_s(p_S^*(a_S))$ (a shock redraw falls below reservation):

$$\left(\nu + (1 - d) \kappa_S(1 - \Gamma_o(p_S^*)) + d f_U \right) u_S = \phi t + \int_{p_S^*}^1 (\xi_S + \delta_S^{\text{end}}) e_S dp, \quad (31)$$

$$(\nu + \lambda_S + \xi_S + s^* \kappa_S(1 - \Gamma_o(p))) e_S(a_U, a_S, p) = \kappa_S u_S \gamma_o(p) + \lambda_S \gamma_s(p) \int_{p_S^*}^1 e_S dp' + \kappa_S \gamma_o(p) \int_{p_S^*}^p s^* e_S dp', \quad (32)$$

The first equation balances the skilled unemployed of a type: they leave by exiting at ν , or by taking a job — in the skilled market at rate $\kappa_S(1 - \Gamma_o(p_S^*))$ when they search there, since an offer is accepted only if it clears reservation, or in the unskilled market at f_U when they have crossed — and they arrive from training completion, at ϕt , or from a destroyed skilled match, exogenously or through a below-reservation redraw. The second balances the employed at each quality p : they leave that quality by exiting, by being shocked away from it at λ_S , by exogenous destruction at ξ_S , or, when searching on the job, by being poached at $\kappa_S(1 - \Gamma_o(p))$; and they arrive there in three ways — a new hire out of unemployment drawing p , a shock redrawing an existing match onto p , and a poached worker moving up onto p from any lower quality. The hire and poaching inflows deposit mass according to the offer density γ_o ; the shock inflow deposits according to the shock density γ_s , which vanishes above δ . for $d = 0$ types, and $e_S \equiv 0$, $u_S = \phi t / (\nu + f_U)$ for $d = 1$ types (consistent with (17)).

10 Skilled-block equilibrium and algorithm

Given $m_S(\cdot)$, f_U , and $E_U(\cdot, 1)$ from the unskilled block, a *skilled-block equilibrium*

$$\{\theta_S, U_S, E_S^0, E_S^1, J_S^0, J_S^1, w_S^0, w_S^1, d, p_S^*, p_S^{\text{oj}}, s^*, u_S, e_S\}$$

satisfies the discrete-choice envelope (21), the employed HJBs (22)–(25), the OJS and separation cutoffs, Nash wages (28)–(29), free entry (30), and the stationary distributions (31)–(32).

10.1 Nested fixed-point algorithm (skilled side)

Iterate on $(\theta_S, p_S^*(a_S), p_S^{\text{oj}}(a_S))$ with $f_U, E_U(\cdot, 1), m_S$ fixed. Inner loop (surplus iteration on the a_S -grid): compute the surplus integrals against each distribution — offer on $[p_S^*, 1]$, shock on $[p_S^*, \delta]$; the two candidate values $U_S^{(0)}(a_S)$, $U_S^{(1)}(a_U)$ and the directed-search policy d via (21); the surplus on the p -grid from (26)–(27); and the Nash split. d is a by-product of the value iteration and needs no separate loop. Outer updates: relocate p_S^*, p_S^{oj} at the sign changes of S_S^{max} and $S_S^1 - S_S^0$; solve the distributions (31)–(32) branching on d ; update θ_S from (30) with Anderson acceleration. On convergence, the skilled block exposes $U_S(\cdot)$ and du_S to the unskilled block; the unskilled block exposes $f_U, E_U(\cdot, 1), m_S$.

11 The offer/shock gap, the on-the-job-search ladder, and endogenous separation

This section develops the economics behind the two skilled-quality distributions of (18). The result is that a single scale parameter δ — the ratio of shock support to offer support — jointly activates two skilled margins that the single-distribution block cannot: the on-the-job-search ladder (hence job-to-job flows) and endogenous separation. The mechanism is the one in Pissarides (2000, Ch. 4, §4.2), transcribed into the RoySearch skilled block.

11.1 Why a single distribution shuts both margins off

In the single-distribution block ($\delta \rightarrow 1$, so $\Gamma_s = \Gamma_o \equiv \Gamma$), the quality a worker draws when *shocked* is distributed identically to the quality drawn from a *fresh offer*. Two collapses follow.

The ladder cannot switch on. On-the-job search pays only if the value of a new offer exceeds the value of the current match by enough to cover the turnover wedge. The gain from search is governed by where a fresh offer lands *relative to where the worker currently sits*. When the worker's current position is itself a draw from Γ and a fresh offer is another draw from the same Γ , there is nothing systematic to climb toward: the expected gain from a search draw does not rise as the current match deteriorates, because a deteriorated match *is* a low draw from the very distribution the offer comes from. Formally, the search cutoff $p_S^{\text{oj}}(a_S)$ solves $S_S^1 = S_S^0$; with $\Gamma_s = \Gamma_o$ the difference $S_S^1 - S_S^0$ carries only the turnover-wedge cost and never the offsetting gain, so it is negative at every p and the cutoff pins to the corner. No employed worker searches; `ee_rate_S` = 0 identically.

Separation cannot switch on. Endogenous destruction requires that being shocked is *bad* — that a redraw can leave the worker worse than her outside option, $S_S(a_S, p) < 0$. But if the redraw is distributed like a fresh offer (mean well inside $[0, 1]$), the redraw option value $\lambda_S \int_{p_S^*}^{\delta} S_S d\Gamma_s$ is large — a shock is nearly as good as a new hire. It swamps the outside-option flow $(r + \nu)U_S$, the reservation numerator stays positive, and $p_S^*(a_S) \rightarrow 0$: no draw is bad enough to abandon. The entire skilled separation hazard is then forced onto the exogenous knob ξ_S .

Both collapses trace to the same source — *offer and shock read the same distribution* — and both are what the estimated single-distribution block exhibits: `ee_rate_S` sits at zero (a moment worth the majority of the objective), p_S^* sits at the corner, and ξ_S carries all separation.

11.2 What the gap supplies

Compressing the shock support to $[0, \delta]$, $\delta < 1$, breaks the symmetry in exactly the way Pissarides (2000) obtains it by placing new matches at the top of the quality support. The offer distribution Γ_o stochastically dominates the shock distribution Γ_s : a shocked worker is *systematically demoted* into $[0, \delta]$, and a subsequent search offer — drawn from $[0, 1]$ — dominates where she now sits. This restores both margins from one primitive.

1. *The ladder switches on.* From a demoted position the expected gain from a fresh offer is positive, so $S_S^1 - S_S^0$ crosses zero at an interior $p_S^{\text{oj}}(a_S) < 1$: low-current-match workers search, high-match workers do not. Job-to-job flows become positive and `ee_rate_S` is matchable.
2. *Separation switches on.* Redraws now land in $[0, \delta]$, so the redraw option value $\lambda_S \int_{p_S^*}^{\delta} S_S d\Gamma_s$ falls. The reservation numerator, the outside-option flow net of that option value, can turn positive; because production is multiplicative in p , a low redraw genuinely destroys output, and the surplus goes negative at low p . The cutoff $p_S^*(a_S)$ lifts off the corner and $\lambda_S \Gamma_s(p_S^*)$ generates separation endogenously.
3. *“Start low, climb” emerges.* New skilled matches form anywhere on $[0, 1]$ and, if low, are climbed away from via on-the-job search toward better matches — a genuine job ladder, not an assumption. Wage dispersion within the skilled sector now has a mechanism: entry dispersion from Γ_o , rungs from the ladder, and demotion spread from the shock.

11.3 Parameterisation and what δ is

The shock is written as $p_{\text{shock}} = \delta p_{\text{offer}}$ with $\delta \in (0, 1]$ — a pure rescaling of the offer draw, so that shape and location vary independently: δ moves only the support, leaving the shared shape

(a_Γ, b_Γ) to govern dispersion. The block is nested: $\delta = 1$ recovers the single-distribution model exactly, and $\delta < 1$ opens the gap.

Two things about δ should be stated plainly, because they are restrictions rather than free outcomes. *The magnitude of the gap emerges; its sign is restricted.* How far shocks demote is estimated (δ is free on $(0, 1]$), and the two skilled flow rates are equilibrium outcomes of $\{\delta, \lambda_S, a_\Gamma, b_\Gamma\}$ — none of the flow *levels* is imposed. But the scaling by $\delta \leq 1$ does fix the *direction*: it places the shock distribution weakly below the offer distribution in the stochastic-dominance order, so a shocked worker is (weakly) demoted, never promoted. This is a deliberate restriction to the economically operative region, not a property that emerges from unrestricted estimation, and it is worth being explicit about why it costs nothing empirically: the opposite direction (shocks that improve matches) produces neither on-the-job search nor endogenous separation — both margins collapse exactly as in the $\delta \rightarrow 1$ case — so it generates zero of the two skilled flow moments the refinement exists to match. Restricting to $\delta \leq 1$ therefore removes only a region the data could never select on these moments; it does not pre-judge a live empirical question. What the estimate of δ then reports is the *size* of the offer/shock gap the data calls for, with $\delta \rightarrow 1$ (no gap, flows collapse) available as the nested alternative the fit can fall back to if the mechanism is not supported.

Regarding *individual* match histories, “start low, climb” is not hardcoded even under this restriction: a worker hired at a high draw does not search and does not climb, while one hired or demoted low does — the trajectory is the output of the reservation and search rules, not an imposed path.

11.4 Relation between the exogenous hazards and the endogenous margin

The exogenous hazard ξ_j and the endogenous margin are complements, not substitutes, and both are estimated. Conditional on the reservation cutoff, ξ_j enters the realised separation hazard *additively and uniformly* — $\xi_j + \lambda_j G(p_j^*)$ — adding the same constant to every match regardless of quality, and (on the skilled side) leaving `ee_rate_S` untouched. The endogenous term $\lambda_j G(p_j^*)$ is *quality-selective*: it destroys only matches whose shocked quality falls below reservation, so it couples `sep_rate` to the offer/shock gap δ , to λ_S , and to the shape (a_Γ, b_Γ) , and it moves with the aggregate state. As *separation events* the two are indistinguishable — both send the worker to unemployment, and under the memoryless redraw neither the separators’ pre-separation wages nor a tenure profile tells them apart — so `sep_rate_j` alone cannot split them. What identifies the split is a *second flow moment that only the redraw produces*: the within-job wage-change hazard among stayers, $\lambda_j [1 - G(p_j^*)]$. A quality shock that lands above reservation is retained but re-prices the match; ξ_j never does this. So the within-job wage-change frequency pins λ_j with no ξ_j in it, and `sep_rate_j` then identifies ξ_j as the residual `sep_rate_j` — $\lambda_j G(p_j^*)$. The reservation p_j^* is separately disciplined by the reservation-driven wage moments (δ , λ_S , the shape moving `sep_rate_S`, `ee_rate_S`, and the skilled wage dispersion together). The endogenous share $1 - \xi_j / \text{sep_rate}_j$ is thus a genuinely targeted object: the corner $p_j^* \rightarrow 0$ that a separation-rate-only fit is free to choose (loading everything onto the flat ξ_j) is ruled out here, because it forces the within-job wage-change hazard to its maximum λ_j — so a measured `wchg_rate_j` strictly below λ_j pushes the estimate off the corner. The within-job wage-change moment is constructed from the SIPP (companion, Part I and the transition-rate identification section); the unskilled market gives the cleanest read because it has no on-the-job search, so a survivor’s wage change is purely the redraw, while the skilled version is informative jointly with the job ladder.

A caution stated plainly, because it bears on the paper’s contribution rather than on the fit.

When the reservation sits at the corner ($p_j^* \rightarrow 0$ — the single-distribution skilled case, and, at the current estimates, the unskilled case as well), the endogenous channel is dead and ξ_j carries the *entire* separation hazard. In that regime ξ_j is doing all the work and the separation moment reduces to “layoffs happened at rate ξ_j ” with no structural content: nothing in it responds to the aggregate state through a mechanism, so it cannot on its own distinguish a demand crisis from a reallocation crisis. The exogenous hazard is included because it is a standard primitive and, disciplined jointly with λ_j by the separation rate and the within-job wage-change moment, it captures the baseline layoff rate honestly; it is *not* a stand-in for the endogenous margin. Whether the endogenous channel activates — whether p_j^* lifts off the corner so that λ_j , δ , and the shock lower tail carry a state-responsive share of separation — remains the open empirical question, and it is the answer to that question, not the size of ξ_j , on which the crisis-classification contribution rests.

12 Observed wages and measurement error

Survey wages are measured with multiplicative log-normal noise, applied after equilibrium and independent of all model objects:

$$\log w_j^{\text{obs}}(\mathbf{a}, p) = \log w_j(\mathbf{a}, p) + \sigma_{w,j} \varepsilon, \quad \varepsilon \sim N(0, 1), \quad j \in \{U, S\}. \quad (33)$$

Because the noise is mean-zero and symmetric, $\mathbb{E}[\log w_j^{\text{obs}}] = \mu_{\log}$, $\text{Var} = \sigma_{\log}^2 + \sigma_{w,j}^2$, $\mu_3 = \mu_{3,\log}$: means, the premium, and third central moments are invariant to $\sigma_{w,j}$; `emp_var_j` is inflated by $\sigma_{w,j}^2$; percentiles are the $\sigma_{w,j}$ -convolved quantiles. $\sigma_{w,j}$ absorbs the residual log-wage dispersion the search model cannot generate at plausible bargaining weights (the Hornstein et al. (2011) gap), is calibrated externally to break the β - σ_w degeneracy, and returns the structural wages at $\sigma_{w,j} = 0$.

Why the matching elasticities and the measurement-error scale are pinned, while the bargaining shares are estimated. Two matching elasticities η_U, η_S and the two measurement-error scales $\sigma_{w,U}, \sigma_{w,S}$ are held fixed; the two bargaining shares β_U, β_S are estimated free and both land interior. The reasons are distinct but linked.

The two *matching elasticities* η_U, η_S are not identified by the moments the model targets. The elasticity of the matching function governs how a meeting rate responds to market tightness, and separating it from the matching efficiency μ_j requires independent variation in tightness — vacancy–unemployment ratios observed across markets or over the cycle within a regime. The estimation targets cross-sectional stocks and flows within a window, which pin the *level* of the meeting rate (through μ_j and the tightness θ_j) but carry no information on its *elasticity*. Both are therefore fixed at $\eta_U = \eta_S = 0.5$, the conventional value in the empirical matching-function literature (Petrongolo and Pissarides, 2001). Lise et al. (2016) fix the same elasticity for the same identification reason.

The two *measurement scales* $\sigma_{w,j}$ are fixed for a separate reason, and fixing them is what *lets* the bargaining shares be estimated. In a Nash-bargained search model the bargaining share and the measurement scale are two ends of one degeneracy on the wage-variance moment: observed dispersion is $\text{Var}(\log w^{\text{obs}}) = \sigma_{\log}^2 + \sigma_w^2$, and that single moment cannot tell a wide structural distribution (β high) from a narrow one seen through noisy data (σ_w high). Exactly one of the pair must be fixed. This model fixes $\sigma_{w,j}$ from an external reliability ratio (below); with the noise scale determined outside the estimation, the wage-variance and higher wage moments

are free to inform β_j rather than being absorbed by a free σ_w . The bargaining shares are then *estimated*, and both come in at interior, economically sensible values — in the base windows $\beta_U \approx 0.19\text{--}0.54$ and $\beta_S \approx 0.32\text{--}0.44$ across the estimated windows, neither railing to a bound. The earlier concern that a free β would rail upward in pursuit of unreachable dispersion does not materialise once σ_w is pinned and the within-job wage-change hazards `wchg_rate_j` are in the moment set: `wchg_rate_S` pins λ_S , which removes the $\beta_S\text{--}\lambda_S$ co-movement that produced the corner in earlier specifications, and β_U is identified through unskilled market tightness (its strongest Jacobian entry is `theta_U`). That the freely-estimated β_U lands near the Lise et al. (2016) lower-education estimate (0.188), from entirely different data and moments, is external corroboration rather than an imposed value.

The measurement scales themselves are set from a reliability ratio rather than fitted. The reliability of a log-wage measure is $\lambda_w = \sigma_{\log}^2 / (\sigma_{\log}^2 + \sigma_w^2)$, the share of observed variance that is signal, and validation studies that match survey reports to employer or administrative payroll records estimate it directly. Bound and Krueger (1991), matching Current Population Survey earnings to Social Security records, report a cross-sectional signal-to-total-variance ratio of 0.82 for men and 0.92 for women. Gottschalk and Huynh (2010), matching the Survey of Income and Program Participation to uncapped W-2 Detailed Earnings Records, report a survey–record correlation of about 0.75 for annual earnings, and Bound et al. (1994) put the reliability of earnings *changes* at roughly 0.75. These figures are all for *annual* earnings, and the same literature is consistent that pay-rate measures are reported substantially less reliably: Duncan and Hill (1985) find hourly earnings less reliably reported than annual earnings, and Rodgers et al. (1993), using employer payroll records, report survey–record correlations of logged values of 0.79 for annual earnings but only 0.60 and 0.46 for the most recent and usual pay period, with the correlation for an hourly rate constructed from reported earnings and hours falling from 0.61 (annual basis) to 0.38 and 0.24 (pay-period basis).

The moment here is the variance of a log real *weekly* wage, a pay-rate concept rather than annual earnings, so the annual-earnings figures are an upper bound on its reliability. Taking the annual estimates as the ceiling and the pay-rate evidence as the reason to sit below it, I set $\lambda_w = 0.78$ — inside the range the literature supports and, relative to the pay-rate-specific estimates, a conservative choice that attributes *less* of the observed variance to noise than a measure-matched reliability would. The value is entered directly, like r , rather than estimated. Inverting it against each window’s observed log-wage variance gives $\sigma_{w,j}$ directly, with no free parameter: $\sigma_{w,j}^2 = (1 - \lambda_w) \text{Var}(\log w_j^{\text{obs}})$, so about 13% of the observed log-wage variance is treated as noise. This is the single source of truth for both scales (the codebase computes them from λ_w at run time), so that a change in the assumed reliability propagates to both markets and all four windows in one place.

13 Estimation moments: model-side construction

The SMM targets the $K = 28$ moments below. Three further quantities are computed but held out of the active objective: the overall unemployment rate `ur_total` (redundant given the two sector-specific rates); the log skill premium `wage_premium`, which is an exact linear combination of `mean_wage_S` and `mean_wage_U` (its Jacobian row equals their difference to numerical precision, so it carries no independent identifying information and would render a full-covariance weight matrix singular); and the skilled EE-move wage step `ee_step_S`, which the memoryless-redraw ladder overshoots by roughly an order of magnitude in every window and is therefore held

out permanently (Section 11). The skill premium remains reported as a fitted quantity; it is simply not a separate target. This section states how each moment is computed *from the solved equilibrium*; the data-side construction and sources are in the companion data-and-moments notes. Notation: aggregates are mass integrals over the type space against the stationary densities, all carrying the copula weight $W_2(a_U, a_S)$. Employed masses are `agg_eU`, `agg_eS`; unemployed `agg_uU`, `agg_uS`; trained stock `agg_t`; the skilled labour force is `agg_mS` = `agg_uS` + `agg_eS`. All wage moments are computed on *log* wages after convolution with the measurement-error noise of (33); the data side matches this by computing on $\log(\text{wage})$.

The objective is, per window,

$$Q(\theta) = [\hat{m} - m(\theta)]' W [\hat{m} - m(\theta)], \quad W = \text{diag}(1/\hat{m}_k^2), \quad (34)$$

i.e. equal weights on the squared *relative* deviation of each moment from its data target $\hat{m}_k$.

Table 1: Model-side construction of the 28 estimation moments. The transition moments reflect the offer/shock refinement of Section 11; the corresponding solver terms are the `equilibrium.jl/skilled.jl` flow-rate computations. The two within-job wage-change hazards `wchg_rate_U/wchg_rate_S` (SIPP) separate λ_j from ξ_j . Three computed quantities are held out of the objective: `ur_total` (redundant), `wage_premium` (an exact linear combination of the two mean log wages), and `ee_step_S` (overshot by the memoryless ladder in every window).

Moment	Block	Model-side computation
<code>ur_U</code>	Stocks	Unskilled unemployment rate <code>agg_uU/(agg_uU + agg_eU)</code> over the unskilled labour force.
<code>ur_S</code>	Stocks	Skilled unemployment rate <code>agg_uS/agg_mS</code> ; set to 1 if the skilled segment is empty.
<code>skilled_share</code>	Stocks	<code>agg_mS/(agg_uU + agg_eU + agg_mS)</code> : skilled labour force over the training-excluded total labour force.
<code>training_share</code>	Stocks	<code>agg_t/total_pop</code> : the stationary trained-in-progress stock t (workers currently in education) as a share of the working-age population.
<code>emp_var_U</code>	Wage shape	Variance of log wages over the employed-unskilled wage density <code>dens_U</code> : $v_0 + \sigma_{w,U}^2$, where $v_0 = \sum_k (\log w_k - \mu)^2 \pi_k$ on the wage grid <code>wmid</code> with normalised mass π_k , and $\sigma_{w,U}^2$ is the additive measurement-error variance from (33).
<code>emp_cm3_U</code>	Wage shape	Third central moment of log wages over <code>dens_U</code> : $\sum_k (\log w_k - \mu)^3 \pi_k$. Invariant to symmetric measurement error.
<code>emp_var_S</code>	Wage shape	As <code>emp_var_U</code> , over the employed-skilled density <code>dens_S</code> ; includes $\sigma_{w,S}^2$.
<code>emp_cm3_S</code>	Wage shape	Third central moment of log wages over <code>dens_S</code> .
<code>jfr_U</code>	Trans	Unskilled job-finding rate $f_U = \theta_U q_U(\theta_U)$, the equilibrium meeting rate for the unskilled unemployed.

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Table 1 — *continued*

Moment	Block	Model-side computation
sep_rate_U	Trans	Employed-weighted unskilled separation hazard $\sum_{a_U} \delta_U(a_U) e_U(a_U) / \text{agg_eU}$, with $\delta_U(a_U) = \xi_U + \lambda_U p_U^*(a_U)^{\alpha_U}$ (exogenous baseline layoff plus the probability a damage shock pushes the match below reservation).
jfr_S	Trans	Skilled job-finding rate, averaged over the skilled-unemployed composition: $d=0$ types accept skilled offers at $\kappa_S(1 - \Gamma_o(p_S^*))$, $d=1$ types accept unskilled offers at f_U ; $f_S = (\text{hire}_S + \text{hire}_X) / \text{uS_mass}$.
sep_rate_S	Trans	Employed-weighted skilled separation hazard $\sum_{a_S} \delta_S(a_S) e_S(a_S) / \text{agg_eS}$, with $\delta_S(a_S) = \xi_S + \lambda_S \Gamma_s(p_S^*(a_S))$ (exogenous layoff plus the probability a shock redraw — from the shock distribution Γ_s — lands below reservation).
ee_rate_S	Trans	Employed-weighted skilled employer-to-employer rate. Only employed skilled with current match $p < p_S^{\text{oj}}(a_S)$ search; each poaches at rate $\kappa_S(1 - \Gamma_o(p_S^{\text{oj}}))$. $\text{ee_rate_S} = \sum e_S(a_S, p) \mathbf{1}\{p < p_S^{\text{oj}}\} \kappa_S(1 - \Gamma_o(p_S^{\text{oj}})) / \text{agg_eS}$.
wchg_rate_U	Trans	Within-job wage-change hazard among unskilled stayers: a λ_U redraw landing <i>at or above</i> reservation survives the match but re-prices it, at rate $\lambda_U(1 - p_U^*(a_U)^{\alpha_U})$, employed-weighted $\sum_{a_U} \lambda_U(1 - p_U^{\alpha_U}) e_U(a_U) / \text{agg_eU}$. Loads on λ_U blind to ξ_U (an exogenous layoff never re-prices a survivor), so it separates λ_U from ξ_U (SIPP; data-and-moments §sipp).
ramsisdec96 wchg_rate_S	Trans	As wchg_rate_U on the skilled side, reading the shock CDF at reservation: $\sum_{a_S} \lambda_S(1 - \Gamma_s(p_S^*(a_S))) e_S(a_S) / \text{agg_eS}$. Separates λ_S from ξ_S .
ee_step_S	Trans	(<i>computed, held out permanently.</i>) Mass-weighted mean log-wage jump of skilled EE moves: $\sum \text{flow}(a_S, p \rightarrow p') [\log w(a_S, p') - \log w(a_S, p)] / \sum \text{flow}$, over poaching moves from $p < p_S^{\text{oj}}$ to an accepted offer p' . The memoryless-redraw ladder produces a step roughly an order of magnitude above the data target in every window (Section 11), so it is reported as a diagnostic but never scored.
mean_wage_U	Wages	Mean log wage over dens_U (measurement error is mean-zero, so unaffected).
mean_wage_S	Wages	Mean log wage over dens_S.
p25_wage_U	Wages	25th percentile of the $\sigma_{w,U}$ -convolved log-wage distribution over dens_U (inverting the noise-convolved CDF).
p25_wage_S	Wages	25th percentile of the $\sigma_{w,S}$ -convolved log-wage distribution over dens_S.
p50_wage_U	Wages	Median of the convolved log-wage distribution, dens_U.
p50_wage_S	Wages	Median of the convolved log-wage distribution, dens_S.
p75_wage_U	Wages	75th percentile of the convolved log-wage distribution, dens_U.

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Table 1 — *continued*

Moment	Block	Model-side computation
p75_wage_S	Wages	75th percentile of the convolved log-wage distribution, dens_S.
theta_U	Tight	Unskilled market tightness θ_U (vacancy–unemployment ratio) from the unskilled free-entry condition.
theta_S	Tight	Skilled market tightness θ_S from the skilled free-entry condition (30).
overlap_UgtS	Overlap	Cross-market wage overlap $\Pr(\log w_U > \text{med } \log w_S) = 1 - F_U^{\text{conv}}(\text{p50_wage_S})$: the $\sigma_{w,U}$ -convolved unskilled log-wage CDF evaluated at the skilled median.
overlap_SltU	Overlap	$\Pr(\log w_S < \text{med } \log w_U) = F_S^{\text{conv}}(\text{p50_wage_U})$: the $\sigma_{w,S}$ -convolved skilled log-wage CDF evaluated at the unskilled median.
ltu_share_S	Duration	Skilled long-term-unemployment share: the u_S -weighted survivor of the skilled unemployment spell past $a^* = 27 \cdot 12/52 \approx 6.23$ months, $\sum_{a_S} u_S(a_S) e^{-\delta_S^{\text{unemp}}(a_S) a^*} / \text{uS_mass}$, with exit hazard $\delta_S^{\text{unemp}}(a_S) = \kappa_S(1 - \Gamma_o(p_S^*(a_S))) + \nu$ (job-finding, at the offer distribution, plus demographic exit).

Remark 3 (Where the offer/shock distinction enters the moments). *Under the offer/shock refinement of Section 11, three transition moments carry the split explicitly. `sep_rate_S` reads the shock CDF Γ_s (a demoting redraw destroys the match); `ee_rate_S` and the job-finding/duration terms read the offer CDF Γ_o (a fresh meeting). In the single-distribution limit $\delta = 1$ — the case the current solver computes, where hire, redraw and poaching all draw from one Γ — the two CDFs coincide, $p_S^* \rightarrow 0$, and both `sep_rate_S` (endogenous part) and `ee_rate_S` collapse to their degenerate values. That collapse is precisely the failure the refinement removes, and it is why the two flow moments jointly identify the gap δ once the two distributions are separated in the code.*

14 Full model equilibrium and global algorithm

A *full equilibrium* consists of unskilled- and skilled-block objects such that: (F1) the unskilled block satisfies (E1)–(E5) given $U_S(\cdot)$ and the cross-market pool du_S ; (F2) the skilled block satisfies its conditions given $m_S = \phi t / (\nu + d f_U)$, f_U , and $E_U(\cdot, 1)$; (F3) the link variables $U_S, m_S, E_U(\cdot, 1), f_U, du_S$ are consistent across the last global iteration.

The full model is an outer fixed point on the link variables, with (U_S, m_S) Anderson-accelerated. Initialise $du_S \equiv 0$ (first pass = baseline unskilled solve), solve the unskilled block, solve the skilled block, update m_S , write back du_S , and iterate to tolerance.

15 Transition dynamics

An unexpected, permanent parameter change occurs at date T . For $t > T$ all hazards and policies are those of the post-switch parameters. A transition equilibrium is a collection of time paths for value functions, policies (including the frontier $\tau(\cdot, t)$ and the cross-market policy $d(\cdot, t)$),

tightness, and distributions, such that value functions solve the time-dependent HJBs, policies are optimal pointwise, distributions solve the forward laws of motion, free entry holds at each date, and the economy travels from the pre-switch to the post-switch stationary distribution. This is the backward–forward, perfect-foresight MIT-shock computation; the shock is unanticipated so the value functions along the path are clean, and ν is the only renewal rate.

15.1 Laws of motion

Densities are over (a_U, a_S) . Training: $\partial_t t = \tau(\cdot, t)u_U - (\phi + \nu)t$. Untrained unemployment: $\partial_t u_U = \nu\ell + (\lambda_U G(p^*(a_U, t)) + \xi_U)e_U - (f_U(t) + \tau(\cdot, t) + \nu)u_U$. Skilled unemployment carries the total destruction hazard $\xi_S + \lambda_S \Gamma_s(p_S^*(a_S, t))$ into the inflow and branches the outflow on d :

$$\partial_t u_S = \phi t + (\xi_S + \delta_S^{\text{end}})e_S^{\text{tot}} - (\nu + (1 - d)f_S(t)(1 - \Gamma_o(p_S^*)) + d f_U(t))u_S,$$

and the trained-segment mass evolves as $\partial_t m_S = \phi t - \nu m_S - d(\cdot, t)f_U(t)u_S$. The cross-market drain $df_U u_S$ moves mass from the skilled to the unskilled segment (absorbed into e_U), preserving total population; ξ_S reshuffles within the skilled segment and leaves m_S unchanged. Tightness is recomputed from free entry at each date with the augmented unskilled seeker pool and the seeker-corrected skilled pool.

15.2 Numerical algorithm

Compute the pre- and post-switch stationary equilibria; guess tightness paths; backward pass for the HJBs (computing $\tau(\cdot, t)$ and $d(\cdot, t)$ at each date via (21)); forward pass for the laws of motion; update tightness from free entry; iterate to tolerance on the tightness paths.

16 Comparative advantage, the cross-market channel, and b_T discipline

This section makes explicit why the always-priced directed-search choice does not contaminate the stationary moments and does not let the training stipend b_T set the training share — the pathology that motivated the redesign.

16.1 The pathology in the single-ability model

With a single ability, the workers who least want to be skilled were the low- x workers, who are bad at *everything*, including unskilled work. Sitting in training to collect b_T cost them almost nothing in forgone unskilled earnings, so a modest b_T made training a weakly dominant “stipend farm,” and the training share responded to b_T globally rather than at the margin. The estimated value functions displayed this directly: the training value exceeded the unskilled-search value even at the lowest ability.

16.2 Why the pathology disappears with two abilities

Proposition 2 (Steady-state cross-flow vanishes endogenously). *In any stationary equilibrium the skilled-unemployed are those who chose to train, i.e. workers above the frontier (high a_S relative to a_U). Under negative ρ_x such workers typically have low a_U , so the unskilled-search branch $U_S^{(1)}(a_U)$ of (21) runs on their weak ability and is dominated by the skilled branch $U_S^{(0)}(a_S)$.*

Hence $d \approx 0$ in steady state — not because the option is prohibited, but because it is never the best option for the workers who are in that state. The option is fully priced; it is simply not exercised.

Proposition 3 (Transition cross-flow emerges from the moving frontier). *Along the transition the frontier $\mathcal{F}$ moves from its pre-switch to its post-switch position. Workers who trained under the old frontier but lie below the new one are mis-sorted skilled workers whose a_U is, by the off-diagonal geometry, decent; for them the always-priced unskilled branch becomes the higher-value choice, so they re-sort downward upon becoming unemployed. The downward flow appears because the regime change placed different workers in the skilled-unemployed pool — an emergent consequence of the state distribution being off its stationary point, not an activated rule. The flow is identically zero at both stationary endpoints (so it never enters the SMM moments, which are computed at the stationary equilibria) and positive only while the frontier is in motion; its speed is governed by the existing f_U , with no new rate.*

Proposition 4 (b_T discipline through forgone earnings). *The workers tempted to farm the stipend are the low- a_S types; under negative ρ_x these are the high- a_U types, who forgo a good unskilled job by sitting in training. The opportunity cost of the training detour is thus high precisely for the would-be farmers, so b_T cannot lure them on the maintained region: raising b_T shifts the marginal genuine trainee at the frontier without buying an arbitrary training share. The suppression works through the forgone-earnings channel and weakens as b_T grows without bound, where the stipend-farm configuration re-emerges.*

17 Model properties

The properties below describe the equilibrium on the economically relevant region of the parameter space: bounded flow values, negative or mild ability concordance, and productivities that keep both markets active. Outside this region some properties fail by construction rather than by error. A sufficiently large training flow b_T makes training weakly dominant for low- a_S types regardless of ability and returns the model to a stipend farm; a strongly positive ρ_x collapses the two abilities to one index (Proposition 1) and, with the wrong curvature of U_U^{train} relative to U_U^{search} , breaks the single-crossing that makes the training frontier a frontier. These configurations are mathematically admissible but are ruled out on economic grounds and are refuted by the data — they destroy the moments the model is estimated on — so the statements below carry the qualifier “on the maintained region” rather than “for all parameters.”

17.1 Comparative-advantage separation

Proposition 5 (Concordance and market populations). *Holding productivities fixed, lowering the skill-copula concordance ρ_x (toward negative correlation) raises the mass of workers with a strict comparative advantage in exactly one market, and hence raises the equilibrium population and tightness of the market that would otherwise be thin. In the limit $\rho_x \rightarrow 1$ the two abilities coincide and the model reduces to the single-index case (Proposition 1), where both markets can be kept populated only by a productivity-shape device.*

Intuition. This is the Gola (2021) concordance comparative static, carried into a search environment. Negative ρ_x places mass in the off-diagonal corners of $[0, 1]^2$: workers good at unskilled and bad at skilled populate U and stay; workers good at skilled and bad at unskilled train and populate S . Separation is earned by comparative advantage, not by curvature of production; the property survives search frictions and the participation margin.

17.2 Training frontier

Proposition 6 (Upward-sloping training frontier). *Under $c'(a_S) < 0$ and T, U_U^{search} continuous, the net training gain $\Delta_T(a_U, a_S) = -c(a_S) + T(a_S) - U_U^{\text{search}}(a_U)$ is decreasing in a_U and increasing in a_S . The training region $\{\Delta_T \geq 0\}$ is bounded below by the frontier $\mathcal{F}$, along which*

$$\left. \frac{da_S}{da_U} \right|_{\mathcal{F}} = \frac{(U_U^{\text{search}})'(a_U)}{T'(a_S) - c'(a_S)} > 0.$$

Workers train above the frontier (high a_S , low a_U) and stay unskilled below it.

Intuition. Training pays through two a_S -channels — a lower direct cost $c(a_S)$ and a higher skilled option value in $T(a_S) = (b_T + \phi U_S(a_S))/(r + \phi + \nu)$ — and is opposed by the a_U -channel of a better unskilled outside option $U_U^{\text{search}}(a_U)$. The frontier is the locus where these balance; it replaces the scalar cutoff $\bar{x}$ of the single-ability model and collapses to it on the $\rho_x = 1$ diagonal.

17.3 Cross-market search frontier

Proposition 7 (Cross-market frontier). *Since $U_S^{(0)}(a_S)$ is increasing in a_S (through the offer-side surplus integral and the linear-in- a_S skilled surplus) and $U_S^{(1)}(a_U)$ is increasing in a_U , the choice $d = \mathbf{1}\{U_S^{(1)}(a_U) > U_S^{(0)}(a_S)\}$ is a frontier in (a_U, a_S) : $d = 1$ for high a_U / low a_S and $d = 0$ otherwise. A fall in P_S (skilled productivity loss) lowers $U_S^{(0)}$ and enlarges the $d = 1$ region; a fall in f_U shrinks it toward the no-cross-market equilibrium.*

Intuition. A trained worker whose skilled prospects have deteriorated relative to her unskilled aptitude redirects search to the market she is better suited to. Under negative ρ_x this region is nearly empty in steady state (Proposition 2) and opens only along a transition (Proposition 3).

17.4 Separation thresholds

Proposition 8 (Separation cutoffs). *For each a_U there is a unique $p^*(a_U) \in (0, 1)$ with $J_U(a_U, p^*) = 0$, given by (12) and decreasing in a_U . For each a_S there are a separation cutoff $p_S^*(a_S)$ ($S_S(a_S, p_S^*) = 0$) and an OJS cutoff $p_S^{\text{oj}}(a_S) \geq p_S^*(a_S)$, both decreasing in a_S ; for $d = 1$ types they retain their definitions but are not realised in stationarity ($e_S \equiv 0$).*

The realised separation hazards by type are $\xi_U + \lambda_U G(p^*(a_U))$ (unskilled) and $\xi_S + \lambda_S \Gamma_s(p_S^*(a_S))$ (skilled), each a quality-independent baseline plus a quality-selective endogenous term. The endogenous skilled channel fires only when the shock redraw can fall below reservation, i.e. when $p_S^*(a_S) > 0$: with a single distribution ($\delta \rightarrow 1$) redraws are as good as fresh offers, so $p_S^*(a_S) \rightarrow 0$ and ξ_S carries the entire hazard; compressing the shock support to $[0, \delta]$ with $\delta < 1$ makes redraws systematically worse than the outside option for low current matches, lifts $p_S^*(a_S)$ off the corner, and lets $\lambda_S \Gamma_s(p_S^*)$ generate separation on its own (Section 11).

17.5 Skilled unemployment duration

A skilled-unemployed worker leaves unemployment only by accepting an offer or by demographic exit; ν is the only route out of the labour force. The exit hazard is constant within type,

$\delta_S(a_S) = \kappa_S(1 - \Gamma_o(p_S^*(a_S))) + \nu$, so spells are exponential within type and the stock survivor is the u_S -weighted mixture

$$\Pr(\text{spell} > a) = \frac{\iint u_S(a_U, a_S) e^{-\delta_S(a_S)a} da_U da_S}{\iint u_S da_U da_S}.$$

Negative duration dependence comes entirely from dispersion in $\delta_S(a_S)$ (length-biased sampling); this is the counterpart of the skilled long-term-unemployment share `ltu_share_S`, evaluated at $a^* = 27 \cdot 12/52 \approx 6.23$ months.

17.6 Wage structure

Wages (13), (28)–(29) are linear in own-ability output plus a bargaining option value; A scales levels without altering shape, and the measurement layer (33) adds dispersion to observed but not structural wages. Following a crisis in which the $d = 1$ region expands, the unskilled-employed composition shifts toward higher a_U (the absorbed ex-trained workers), so aggregate unskilled wages rise through composition alone.

17.7 Identification of ρ_x and the premium/composition split

Proposition 9 (Separate levers). *The wage premium is set by the level ratio P_S/P_U (a pure level effect that does not touch the training incentive), while the skilled/unskilled composition is set by ρ_x (how separated the two ability pools are). In the single-ability model one shape parameter was forced to hit both, which over-identified it; here the two targets have distinct levers.*

The natural moments for ρ_x are the cross-market wage overlaps (`overlap_UgtS`, `overlap_SltU`): strong negative ρ_x produces cleanly separated wage distributions (little overlap), weak or positive ρ_x produces heavy overlap. Confirming that the overlaps respond to ρ_x independently of P_S/P_U is the first identification check. Because unskilled production is $AP_U a_U p$, strictly increasing in a_U , the unskilled ability now shapes wages, reservation qualities, and separation across its support, so a_ℓ and ρ_x touch the model through many moments rather than a single composition scalar — resolving the under-identification of the type distribution that arose when unskilled production was flat in ability.

17.8 Identification of the offer/shock gap δ

The gap parameter δ and the shock rate λ_S are jointly identified off the two skilled flow moments they generate. The job-to-job rate `ee_rate_S` responds to the width of the search band $[p_S^*, p_S^{\text{oj}}]$, which widens as δ falls (offers dominate demoted matches by more) and as λ_S rises (more workers are demoted into the searching region). The endogenous separation rate `sep_rate_S` responds to the shock lower tail below reservation, $\lambda_S \Gamma_s(p_S^*)$, which likewise rises as δ falls. The two moments pin the pair: a lower δ raises both flows, while λ_S trades their relative magnitude (it scales separation one-for-one but affects job-to-job flows only through the demoted stock), so the ratio `ee_rate_S/sep_rate_S` separates δ from λ_S . The shared shape (a_Γ, b_Γ) is identified off skilled wage dispersion (`emp_var_S` and the skilled percentiles), now that it governs dispersion alone rather than being loaded onto separation and the ladder simultaneously. If the two flow moments cannot be reached jointly, ξ_S is retained to carry residual separation and is identified off `sep_rate_S` at the margin where the endogenous channel saturates.

17.9 Market tightness and the transition

Tightness responds through (i) vacancy costs k_j , scale A , and productivity; (ii) composition of the unemployed pool; (iii) matching technology μ_j, η_j ; (iv) the cross-market channel, where a shift in $d(\cdot)$ inflates one seeker pool and deflates the other; and (v) the skilled hazard ξ_S . The aggregate skilled share satisfies

$$\dot{M}_S(t) = \phi \mathcal{T}(t) - \nu M_S(t) - f_U(t) \iint d(a_U, a_S, t) u_S(a_U, a_S, t) da_U da_S,$$

adding a cross-market drain to demographic turnover and training completion. When the crisis opens the $d = 1$ region, M_S adjusts faster than demographic replacement alone would allow, then reverts to the $\ln 2/\nu$ half-life once the channel closes.

Demand vs. reallocation shock. A demand shock (financial crisis) traces a movement along the Beveridge curve — unemployment rises and tightness falls in both segments — captured by a fall in A (and P_U, P_S), a rise in k_U, k_S and possibly the destruction rates $\lambda_U, \lambda_S, \xi_U, \xi_S$, holding matching technology fixed; the cross-market region stays closed. A reallocation shock (COVID) shifts the Beveridge curve outward — tightness rises despite higher unemployment, separations rise sharply on the skilled side — captured by a fall in μ_U, μ_S , a rise in λ_S (and ξ_S), and a fall in P_S ; the skilled-side separation spike now runs primarily through the endogenous channel, as a higher shock rate λ_S pushes more matches below reservation, and the cross-market region opens so prior trained cohorts re-sort into unskilled jobs. The multi-regime SMM identifies which parameters move in each crisis, and the transition computation traces the implied path of the frontier and tightnesses.

18 Assessment of fit

This section reads the estimated fit as a set of tensions the model must reconcile. The reference point is the base-fc estimate under the shipped inverse-sampling-variance weighting, with both bargaining shares estimated free. The organising question is not whether each residual is zero — with 28 moments and 25 free parameters it cannot be — but *why* a given residual is not zero: which moments compete for the same parameter, and which structural feature forces the trade-off. The misfit is not scattered. It is concentrated entirely in the skilled flow-and-duration block, and the cross-section is matched almost exactly.

18.1 What the model matches

The wage distribution, in both markets and across its whole shape. Every wage-level moment lands within a few tenths of a percent of its target: the two mean log wages, all six quartiles, and (as a derived quantity) the skill premium. Figure 4 overlays the model log-wage densities on kernel density estimates of the data for skilled and unskilled workers; the model reproduces the location and spread of both distributions, not merely their means. The mechanism is direct: the sectoral productivity shifters P_U, P_S set the levels, the ability copula sets the cross-market overlap, and both are strongly identified.

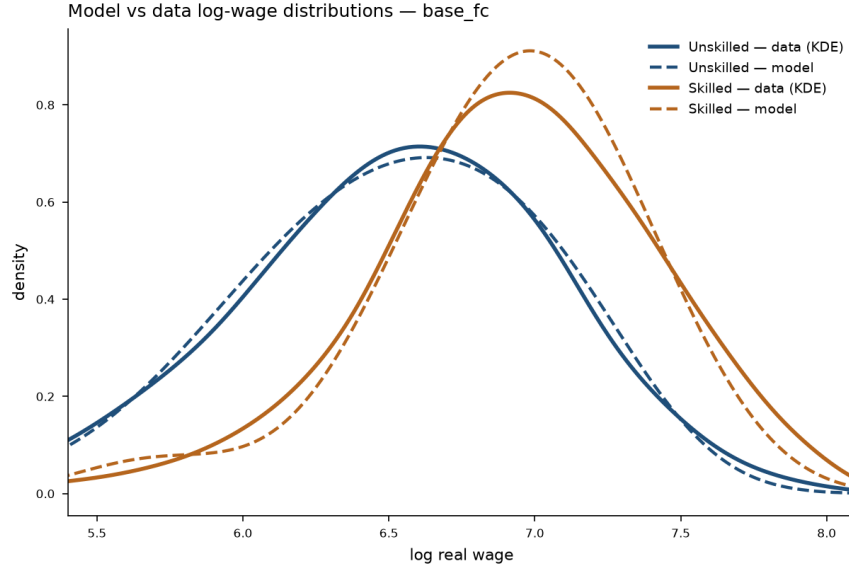


Figure 4: Model versus data log-wage distributions, base-fc. Solid: kernel density estimates of $\log(\text{real wage})$ from CPS ASEC (ASECWT-weighted), skilled and unskilled. Dashed: model log-wage densities, convolved with the calibrated measurement error. The model reproduces the location and spread of both distributions; the skilled model density is slightly too peaked, the shape residual behind the `emp_cm3_S` miss.

Wage dispersion. With both β free and the measurement scale pinned, the employment-weighted log-wage variances are matched — `emp_var_U` to within a few tenths of a percent and `emp_var_S` to within about five percent. This is the reversal of the earlier concern that the model could not generate the observed dispersion: once σ_w is fixed externally, the wage moments identify β , and the estimated β supplies the dispersion. What remains is a shape residual, not a scale one — the skilled third moment `emp_cm3_S` is about twenty percent off, visible in Figure 4 as a slightly too-peaked skilled density.

Unemployment, composition, and the unskilled flows. The two unemployment rates, the skilled employment share, and the training share are all close. And the model’s central mechanism is alive where it should be: on the unskilled side, roughly two-thirds of separations are endogenous (Figure 5), a real interior margin that reaches into the ability mass and moves across windows. This is the object the crisis-classification story rests on, and it works.

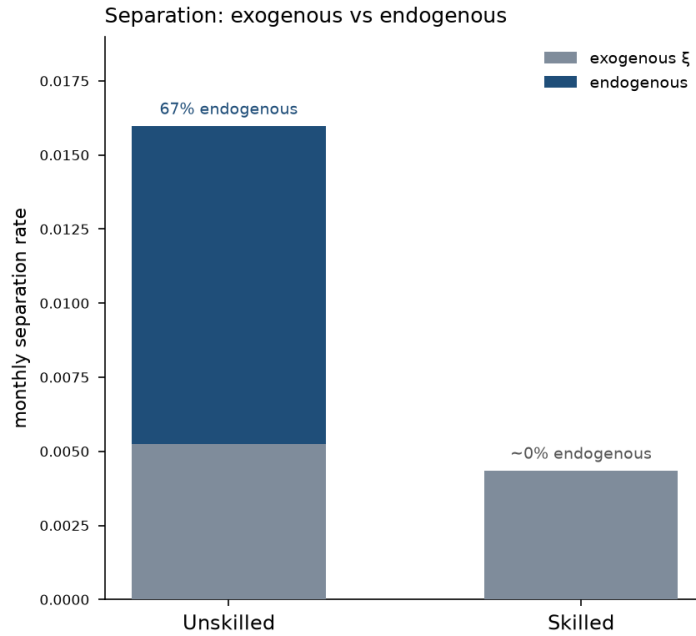


Figure 5: Separation decomposed into exogenous (ξ_j) and endogenous parts, by market. The unskilled margin is a sizeable interior object ($\approx 67\%$ endogenous); skilled separation is essentially all exogenous.

18.2 Where the residual is not zero, and why

Figure 6 orders the moments by their contribution to the objective, and Table 2 gives the underlying targets, model values, and signed deviations. The misfit is concentrated in four skilled-market quantities — `sep_rate_S`, `ltu_share_S`, `jfr_S`, and `ee_rate_S` — and each reflects a tension the model must reconcile rather than a free parameter left unfit.

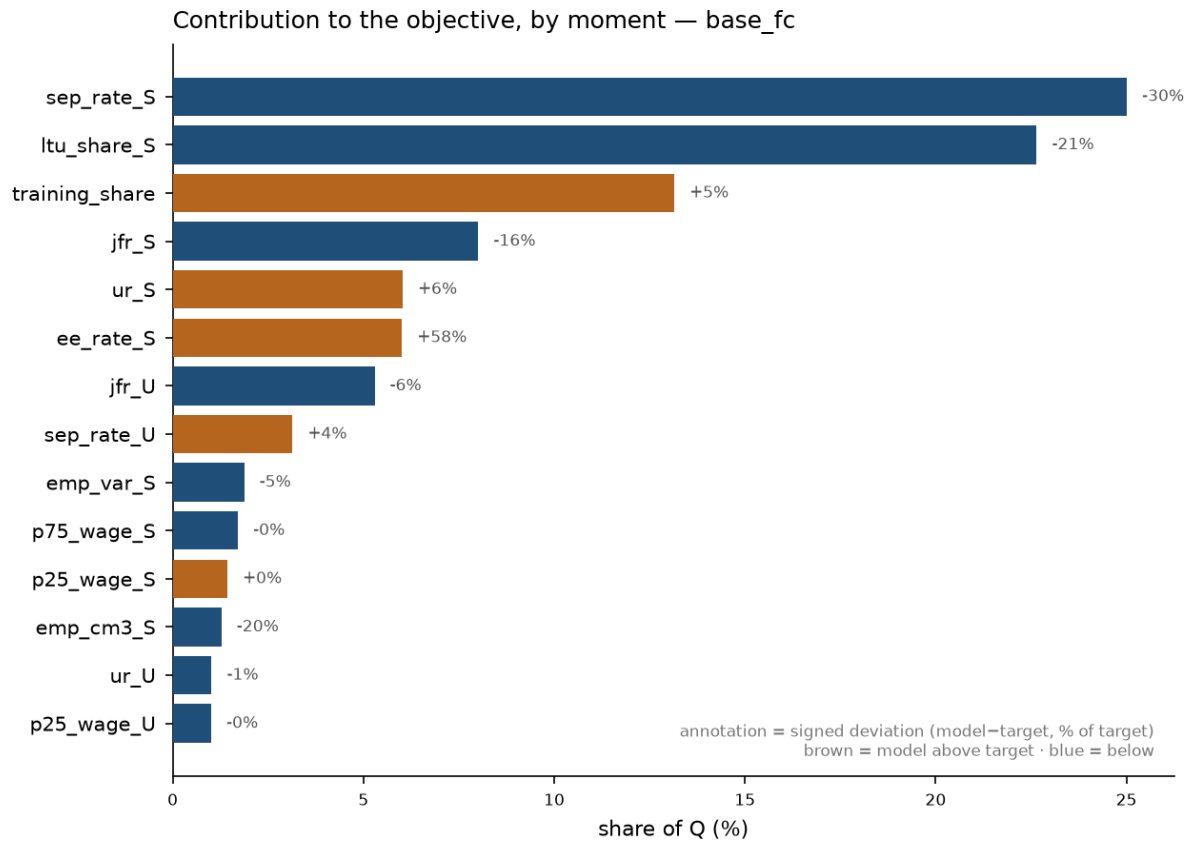


Figure 6: Contribution of each moment to the objective at the base-fc estimate (bar length = share of Q ; annotation = signed relative deviation, model minus target). The misfit is concentrated in the skilled flow block: `sep_rate_S`, `ltu_share_S`, `jfr_S`.

moment	target	model	dev (%)	share of Q (%)
sep_rate_S	0.0062	0.004355	-29.8	25.0
ltu_share_S	0.2326	0.1828	-21.4	22.6
training_share	0.04	0.0419	+4.7	13.1
jfr_S	0.3204	0.2695	-15.9	8.0
ur_S	0.026	0.02752	+5.8	6.0
ee_rate_S	0.008	0.01265	+58.1	6.0
jfr_U	0.3025	0.2859	-5.5	5.3
sep_rate_U	0.0153	0.01598	+4.4	3.1
emp_var_S	0.2461	0.233	-5.3	1.9
p75_wage_S	7.257	7.236	-0.3	1.7
p25_wage_S	6.62	6.639	+0.3	1.4
emp_cm3_S	-0.0572	-0.06869	-20.1	1.3
ur_U	0.0636	0.06296	-1.0	1.0
p25_wage_U	6.176	6.158	-0.3	1.0
p75_wage_U	6.914	6.928	+0.2	0.7
skilled_share	0.3016	0.3008	-0.3	0.6
p50_wage_S	6.936	6.947	+0.2	0.5
mean_wage_S	6.915	6.909	-0.1	0.2
wchg_rate_S	0.1292	0.1334	+3.3	0.1
overlap_SltU	0.2032	0.1993	-1.9	0.1
wchg_rate_U	0.1505	0.1494	-0.8	0.1
overlap_UgtS	0.2395	0.2384	-0.5	0.0
mean_wage_U	6.53	6.529	-0.0	0.0
emp_var_U	0.2967	0.2961	-0.2	0.0
theta_U	0.4346	0.4345	-0.0	0.0
theta_S	1.211	1.211	+0.1	0.0
emp_cm3_U	-0.0397	-0.03984	-0.4	0.0
p50_wage_U	6.556	6.555	-0.0	0.0

Table 2: Per-moment fit at the base-fc estimate (28-moment objective, variance weighting). *dev* is the signed relative deviation $100(\text{model} - \text{target})/|\text{target}|$; *share of Q* is each moment’s contribution to the objective, renormalised over the active set. The misfit is concentrated in the skilled flow-and-duration block (**sep_rate_S**, **ltu_share_S**, **jfr_S**), which together carry over half the objective; every wage-level moment sits within 0.3% of its target. **wage_premium** is omitted (held out as an exact linear combination of the two mean log wages); **ee_step_S** is likewise held out.

The skilled separation rate is below target, and it cannot be raised without breaking skilled unemployment. At the estimate the skilled endogenous margin sits at its floor, so **sep_rate_S** $\approx \xi_S$, and the fitted ξ_S leaves the rate about thirty percent below its data value. The reason it is not raised is a stock–flow identity: every skilled separation must pass through skilled unemployment before re-matching, so raising the separation rate raises the skilled unemployment stock roughly in proportion to $1/(\text{job-finding rate})$. Moving ξ_S up to hit **sep_rate_S** drives **ur_S** far past its target, and because the unemployment rate is measured precisely (small sampling variance, hence large weight), the estimator prefers to miss the separation rate. Sweeps of the four structural levers that could raise the endogenous margin generalise this: every one of them (the skilled outside option b_S , the shock support δ , the offer shape a_F ; the on-the-job-search cost σ_S does not move the margin at all) buys a higher endogenous share only by overshooting **ur_S** or collapsing the wage premium. The point is developed as a general property in Section 19. The

two skilled moments over-determine the one parameter, and the weighting resolves the conflict in favour of the more precisely measured one.

The endogenous skilled margin is near zero, and no primitive turns it on cheaply. One might hope to hit `sep_rate_S` through endogenous separation rather than through ξ_S — lifting the reservation p_S^* into the shock mass so that bad matches dissolve. Direct sweeps of the three structural levers that could do this show each one pays elsewhere: raising the skilled outside option b_S or narrowing the shock support (via δ) turns on separation only by driving the reservation so high that job-finding collapses and `ur_S` explodes; putting more mass on low-quality draws (lowering a_L) does reach the separation target while holding `ur_S` near its value, but it collapses the wage premium, because the low-quality matches it creates are precisely the low-paying ones. The skilled separation rate, unemployment rate, job-finding rate, and wage premium jointly determine the skilled block; the endogenous separation share is whatever those four leave over, and here that is near zero. Skilled separation is therefore exogenous in the estimated model not by assumption but as a consequence of the joint fit — a structural limitation of the memoryless-redraw mechanism, which uses the same quality redraws to hit the wage-change rate and the job ladder and cannot simultaneously use them to generate separation.

The skilled duration distribution has the wrong shape. The skilled long-term-unemployment share is about twenty percent below target while the skilled job-finding rate is also below target. Lower job-finding should lengthen durations and *raise* the long-term share, so the two misses pointing the same way indicate that the dispersion of exit hazards across skilled ability types is too narrow: the model does not generate enough of a long right tail of stuck workers. This is the one genuinely open shape weakness, distinct from the level tensions above, and worth naming as such.

The skilled EE-move rate overshoots but is lightly penalised. The model produces more skilled job-to-job moves than the data. Under inverse-sampling-variance weighting this moment carries little weight (its sampling variance is large), so the estimator tolerates the overshoot in exchange for the tightly measured rate fits — the weighting behaving as designed, reported here for completeness rather than as a failure. The EE *wage step*, `ee_step_S`, remains held out of the objective throughout, as the memoryless ladder overshoots it by roughly an order of magnitude in every window.

18.3 Reading

The through-line is that the model fits everything about the cross-section — wage levels, the wage distribution, dispersion, unemployment stocks, and composition — together with the unskilled flows, and its misses are confined to the skilled flow-and-duration block, where a small number of precisely measured moments over-determine the market. The central asymmetry that results — endogenous separation on the unskilled side, exogenous separation on the skilled — is not imposed; it is what the joint identification delivers, and it is a statement about how the two markets differ in why jobs end.

19 A general tension: the skilled market resists being made worse

The largest misses in the fit all point the same way, and they are one fact, not several. The model under-produces the skilled separation rate (`sep_rate_S`, -30%), the skilled long-term-

unemployment share (`ltu_share_S`, -21%), and the skilled job-finding rate (`jfr_S`, -16%); and no admissible parameter move raises skilled separation without either overshooting the skilled unemployment rate or collapsing the wage premium (Section 18). In each case the model wants a skilled market that is *healthier* than the data — more durable matches, fewer separations — and resists being pushed toward the worse market the data describe. This is not a collection of independent shortfalls; it is a single structural property surfacing in several moments.

The property. The skilled match surplus is large and positive over almost the entire type-quality space. Two consequences follow directly, and they are the two places the model has needed a countervailing device.

First, it is why separation requires the copula. If the skilled surplus is positive for nearly every (a_S, p) , then almost every worker with any skilled aptitude strictly prefers a skilled match to unskilled search, and in an unconstrained model everyone would train. The unskilled market would empty. Something must hold workers out of skilled employment, and in this model that something is the negative ability correlation $\rho_x < 0$ (Proposition 5): a worker with high skilled aptitude tends to have low unskilled aptitude and vice versa, so the comparative-advantage margin — not a wedge in the level of skilled value — is what keeps both markets populated. The separation of the two markets is manufactured against a skilled market that is otherwise too attractive to leave.

Second, it is why skilled separation is exogenous. The same positive surplus means the endogenous quit/destruction margin $p_S^*(a_S)$ sits at its floor for essentially all a_S : even a low-quality redraw leaves the match worth keeping, so no shock short of an exogenous layoff dissolves it. Skilled separation is therefore all ξ_S , the endogenous share is near zero (Section 11), and `sep_rate_S` cannot be lifted by any device that works through the reservation — raising the outside option b_S , narrowing the shock support δ , or fattening the low-quality tail a_Γ — without side effects that a targeted moment rejects. The stock-flow identity does the rest: any separation the model does produce must pass through skilled unemployment, so forcing more of it drives `ur_S` above target, which under variance weighting is expensive.

Why this is the right way to read the misfit. The skilled market in the data separates more, finds jobs a little less readily, and carries a longer unemployment tail than a frictional market with surplus this high and this uniform can generate. The model does not have a free parameter that makes skilled employment fragile while leaving the wage premium, the composition, and the unemployment rate intact, because fragility and value are the same object: the surplus that pays the premium and pulls workers into training is the surplus that keeps their matches alive. This is the manual-calibration difficulty stated structurally — the skilled unemployment rate could not be brought down (or, equivalently, skilled separation brought up) without disturbing many other moments, because a single primitive property governs all of them.

19.1 The reservation sits at the floor, and a surplus condition says why

The endogenous quit/destruction cutoff $p_S^*(a_S)$ is the match quality at which the maximised skilled surplus $S_S(a_S, p) = \max\{S_S^0, S_S^1\}$ crosses zero: matches below it dissolve, matches above it survive. At the base-fc estimate $p_S^*(a_S) = 0$ for every ability type that carries skilled employment. Weighting by the skilled employment mass, *the entire skilled workforce sits at the floor* — 100% of employed skilled mass has $p_S^* = 0$. The cutoff is interior only for the lowest skilled-aptitude types (the bottom quarter of the a_S grid), and under the negative ability correlation $\rho_x < 0$

those are exactly the types that do not select into skilled employment: they have high unskilled aptitude, direct their search to the unskilled market, and never form a skilled match whose quality could fall below a reservation. The reservation is therefore at the floor precisely where it matters and interior only where no mass lives.

The condition behind this is a surplus inequality. $p_S^*(a_S) = 0$ exactly when the maximised surplus is already non-negative at the *worst possible draw* — the lowest point of the quality grid:

$$S_S(a_S, p_{\min}) \geq 0.$$

The worst draw is worth keeping, so no realised shock can push the match below a reservation that does not bind. Numerically at the estimate this holds with room to spare over the employed range: the worst-draw surplus $S_S(a_S, p_{\min})$ is strictly positive for every ability type at the floor (median value of order 10^3 in model flow-utility units, rising with a_S), and turns non-positive only for the lowest-aptitude types, which is where and why the cutoff becomes interior. The distance by which $S_S(a_S, p_{\min})$ clears zero over the skilled range is the quantitative content of “the market resists being made worse”: to generate endogenous separation one must push that surplus down through zero, and at the estimate it is far above.

19.2 No parameter buys endogenous separation without paying through a targeted moment

The resistance can be read parameter by parameter, treating each moment as the *channel* through which a lever’s cost is paid rather than as a target to be hit. Starting from the estimate (skilled endogenous separation share 0.13%, $\text{ur_S} = 2.75\%$), the three structural levers that can raise the endogenous share each move it only by moving a moment the fit needs:

lever	$\Delta(\text{endog. share})$	$\Delta \text{ur_S}$	$\Delta(\text{premium})$	channel that pays
$b_S \uparrow$ (outside option)	+0.33pp	+0.25pp	+	ur_S (stock–flow)
$\delta \downarrow$ (shock support)	+0.03pp	+0.19pp	–	ur_S (stock–flow)
$a_\Gamma \downarrow$ (offer shape)	+3.95pp	–0.01pp	–	the wage premium

The reading is uniform. Raising the skilled outside option b_S or narrowing the shock support δ lifts the reservation, which does turn on some endogenous separation, but every separation must flow through skilled unemployment, so ur_S rises with it — for b_S , about 0.76 points of skilled unemployment per point of endogenous share, and for δ the trade is far worse, roughly 7 points of ur_S per point of endogenous share, because δ moves the reservation very little for a large disturbance to the quality distribution. Fattening the low-quality tail of the offer distribution ($a_\Gamma \downarrow$) is the only lever that generates a sizeable endogenous share while holding ur_S nearly fixed — but it does so by manufacturing low-quality matches, which are the low-paying ones, and the wage premium collapses (the largest premium loss of the three). There is no lever whose channel is free: the skilled unemployment rate and the wage premium are precisely the moments the estimate fits well, and each lever spends one of them to buy separation the data are asking for. This is the same surplus property of the previous subsection seen from the parameter side — the primitives that would fragilise skilled matches are the primitives that set skilled unemployment and skilled pay, so they cannot be moved independently of them.

20 Open items

These are flagged as work to complete once the two-ability solver is running; none is assumed solved.

1. **Unskilled wage dispersion.** Whether the two-ability structure lifts `emp_var_U` to target is empirical; the damage shock alone bounds structural variance below the target (the ceiling $\text{Var}(\log w_U) = \beta_U^2 (AP_U)^2 \text{Var}(p)$ with $p \in [0, 1]$ and small β_U).

Resolved. Both bargaining shares are now estimated free, and both land interior ($\beta_U \approx 0.19\text{--}0.54$, $\beta_S \approx 0.32\text{--}0.44$ across the base windows). Under the shipped inverse-sampling-variance weighting the dispersion moments are matched: at the base-fc estimate `emp_var_U` is within 0.2% of target and `emp_var_S` within 5%. The β_U^2 ceiling concern is retired — freeing β_U off the wage moments (rather than pinning it) is precisely what supplies the unskilled dispersion, and it does so without β_U railing, because σ_w is fixed externally and unskilled market tightness identifies β_U independently of the variance moment. The only residual dispersion miss is the skilled third moment `emp_cm3_S` ($\approx -20\%$), a shape rather than a scale gap.

2. **b_T discipline.** Verify Proposition 4 numerically by sweeping b_T and confirming `training_share` no longer runs away.

Resolved. The b_T farm is no longer the best choice for disciplining the training share, because the copula now does that work: comparative advantage across the two abilities determines who trains, so the training margin is pinned by the joint ability distribution rather than by the training-flow value alone. Education remains expensive where it should be — for workers with low skilled aptitude a_S , whose training cost is high — so the discipline comes from the cost of skill acquisition given ability, not from a runaway b_T .

3. **Symmetric vs. asymmetric marginals.** Start symmetric; move to per-axis marginals only if the overlaps cannot be fit.

Resolved. The marginals are identical. The correlation ρ_x alone was enough to fit the overlaps, so there is no need to introduce per-axis marginals: the model uses no more parameters than it absolutely needs, and the single concordance parameter carries the separation.

4. **Skilled-side flows and dispersion.** The offer/shock refinement (Section 11) is expected to activate `ee_rate_S` and endogenous `sep_rate_S` jointly and to relieve the pressure on `emp_var_S`. Whether the fitted equilibrium reaches all three, and whether it does so without ξ_S , is an estimation question settled by the solved fit, not assumed here: the prototype at $\delta = 1$ must reproduce the single-distribution run before any $\delta < 1$ result is trusted.

Resolved. The reproduction gate passed: at $\delta = 1$ the offer/shock block reproduces the single-distribution run to numerical tolerance, and the estimated $\delta < 1$ specification activates `ee_rate_S`. Two facts settle the item. (i) Skilled separation is essentially all exogenous at the estimate: the endogenous margin sits at its floor ($p_S^* \approx 0$), so `sep_rate_S` $\approx \xi_S$. This is forced by the joint fit rather than assumed — the skilled separation rate, unemployment rate, job-finding rate, and wage premium jointly over-determine the skilled block, and the endogenous share is the near-zero residual. Direct sweeps confirm that every structural lever which raises the endogenous margin (b_S , δ , a_T) does so only by overshooting `ur_S` through the stock-flow identity or by collapsing the wage premium (Section 18). (ii)

With both β estimated free and the shipped inverse-sampling-variance weighting, `emp_var_S` is matched to within 5%, so the dispersion pressure is relieved directly rather than dumped into $\sigma_{w,S}$. The one moment the δ ladder cannot reach is the EE wage step `ee_step_S`, held out of the objective (Section 11).

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